

Grandfathering at Sale: Nonportable School Rights and Housing Exchange

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Abstract

When a public entitlement is grandfathered only while an asset is held, selling forfeits what holding preserves. I study Beijing's January 2019 title-vintage rule, under which transfers moved buyers from a single-school mapping to multi-school assignment. In 54 exactly linked communities, formerly key and ordinary communities both begin near three platform completions per 1,000 Lianjia-recorded housing units. Over the next sixteen weeks, ordinary communities rise to about six while formerly key communities remain near three. The raw exposed-relative PPML ratio is 0.57–0.61; seasonal benchmarks imply 0.42–0.67, with the largest gap against an unusually strong 2018. The negative sign survives frozen samples, homogeneous completion clocks, count estimators, and every school and community deletion. Public active-stock and asking-price comparisons move less, conditional on unverified crawler comparability, but do not identify a listing hazard. A search-and-bargaining model shows how a sale-triggered entitlement reset can remove marginal matches without requiring a commensurate selected-price response.

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1 Introduction

Suppose keeping a house preserves a valuable public-school entitlement, but selling creates a title to which that right does not travel. What gives—the property’s price, or the buyer–seller match? A price can divide surplus between the two sides; it cannot restore a right destroyed at transfer.

Grandfathering can add an extensive-margin selection response to ordinary capitalization. Price still divides surplus conditional on trade. But when transfer destroys part of the bundle, a lower price can shift surplus between buyer and seller without restoring what was lost; for some pairs no mutually acceptable price remains. The title rule can therefore change which buyer–seller matches complete even when prices among surviving trades move comparatively little.

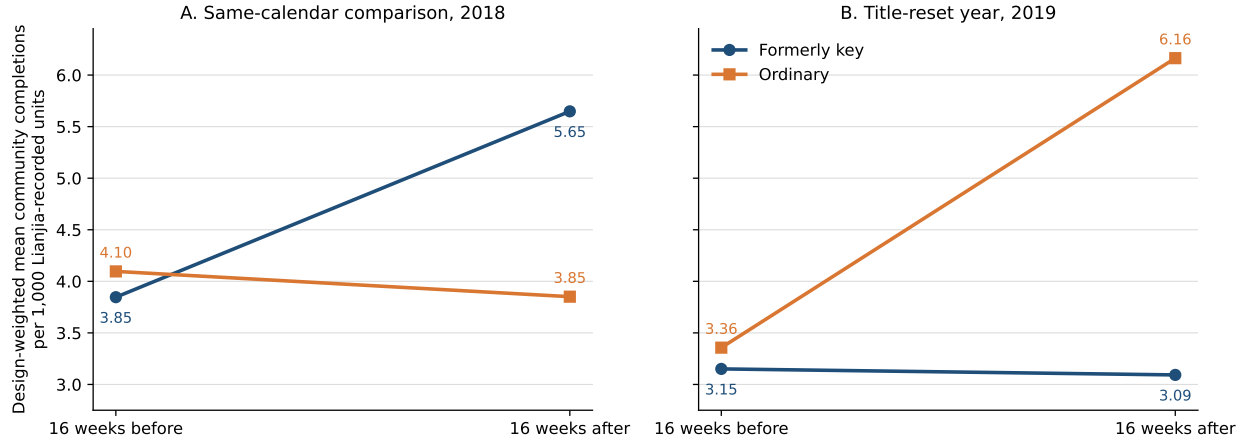
Grandfathering thus solves one problem and can create another. It protects people who relied on an old rule, but protection that survives only while an asset is retained makes exchange the forfeiture event. The stakes can be large: using a representative nationwide dataset, [Batzet et al. \(2026\)](#) estimate that mortgage-rate lock-in was associated with 1.72 million fewer arm’s-length transactions between 2022Q2 and 2024Q2. California’s property-tax portability reforms show the reverse margin, raising mobility among eligible older homeowners ([Ferreira, 2010](#)). The same logic applies whenever a favorable mortgage, property-tax basis, license, or public benefit is tied to the incumbent rather than transferable with the asset ([Fonseca and Liu, 2024](#)). I ask what happens when a public right disappears precisely because its owner trades—and what object a transition right should follow.

Beijing’s Haidian district provides a sharp setting. Residential addresses historically mapped eligible households to particular public primary schools. A reform announced in 2018 preserved the prior single-school rule for titles registered before January 1, 2019 while they remained under the old title. A buyer whose acquisition generated a post-cutoff title instead entered multi-school assignment ([Haidian District Education Commission, 2018a,b](#)). The dwelling did not change at transfer; the school-assignment regime obtained through purchase did. The policy supplies a legal date, a pre-policy map of differential exposure, and a direct reason why the seller’s retained bundle differs from the buyer’s acquired bundle.

The central empirical fact is easy to state after putting communities on a common recorded-unit scale: *three stays three while three becomes six*. In the sixteen weeks before January 1, 2019, both groups average roughly three platform completions per 1,000 Lianjia-recorded housing units. Over the next sixteen weeks, ordinary-school communities rise to about six while formerly key-school communities remain near three. [Figure 1](#) shows the policy-year divergence and the same-calendar comparison one year earlier.

The policy changed both groups, so ordinary-school communities are not an untreated control. The design instead measures differential policy incidence: replacing a prior key-school mapping is more consequential than replacing an ordinary-school mapping. A seasonal triple difference compares the formerly-key–ordinary winter-to-spring differential in 2019 with the same calendar differential in earlier years. The frozen frame contains fifty-nine

Figure 1. The missing New Year rebound on a common recorded-unit scale



Notes: Each point is the design-weighted mean of community-level platform completions per 1,000 Lianjia-recorded housing units on the 54-community exact-ID public-data support, using the lower completion-date assignment. The upper assignment gives the same qualitative pattern. The denominator is November-2019 Lianjia page metadata, not contemporaneous or cadastral stock; the vertical axis is truncated. Panel A is the same sixteen-week calendar comparison one year before implementation. This scale check does not replace the frozen count design.

notice-verified communities, nineteen historically assigned to seven schools classified as key, 7,340 completed transactions, and 1,173 transactions in the designated 2018–2019 cells.

I establish the argument in four steps. First, formerly key-school communities do not join the 2019 winter-to-spring expansion observed in Beijing outside Haidian, other Haidian communities, and the matched ordinary group. Before subtracting any prior year, the exposed-relative PPML ratio is 0.57–0.61. Same-calendar seasonal benchmarks imply ratios from 0.42 to 0.67; the designated 2018 comparison produces the largest gap because the 2018 exposed-relative seasonal change is unusually positive. In an exhaustive fixed-weight label-sensitivity exercise, the raw 2019 grouping lies in the most negative 2.1–2.2 percent of all 116,280 seven-of-twenty-one school labelings one-sided. This is not randomization inference, because exposure and the matched frame were not randomly assigned.

Second, I ask at which observed market stage the gap appears. On the late-March public-stock support, the active-stock ratio is 0.923 while same-support completed-flow ratios are 0.368 and 0.427; equality tests give $p = .029/.033$ and a two-baseline Holm value of .066. A November comparison gives the same ordering with weaker inference. In a separate exact-ID link from November 2017 to November 2019, active listings rise in both groups with no detectable differential change. Available asking-price contrasts are small and imprecise. These comparisons are conditional on unverified treatment-relative crawler coverage and do not observe entry, withdrawal, relisting, or conversion hazards. They point toward transaction execution as an adjustment margin; they do not identify delay, buyer composition, match quality, or a unique primitive.

Third, I develop a search-and-bargaining model in which the title reset raises the bilateral match threshold. Conditional on match opportunities, a higher threshold removes marginal

matches and changes selection into observed prices. The completed-flow response is signed; the conditional-price response is not. The model supplies one mechanism that can rationalize a large relative-flow response alongside small, imprecise selected-price movements; the data do not identify that mechanism uniquely.

Fourth, I use the same primitives to compare four transition rules. Protection can follow the standing title, the property, the household, or a common calendar date. When the prior mapping is at least as valuable as the expected post-transfer bundle, property portability, household portability, and a uniform transition each weakly expand the bilateral match set relative to title-contingent grandfathering. This is a partial-equilibrium match-set ordering, not a welfare ranking.

The analysis keeps three observation universes separate. A 586,312-transaction platform archive measures completed exits; posting and completion clocks provide accounting within the selected set observed to complete by the archive end; and public crawler snapshots measure active offers on fixed supports. Historical admission notices reconstruct pre-policy school assignments without using transaction outcomes, and a fixed one-day interval bounds missing completion dates. Re-dating every transaction with a uniform posting-plus-duration clock leaves the core estimates nearly unchanged. Exact-day observability nevertheless changes differentially across the headline cells, so treatment-neutral platform capture remains a necessary maintained condition. The online appendix reports this threat, every failed mechanism family, and the complete protocol history.

Related literature. I make three contributions. First, school-capitalization studies examine how access enters property values (Black, 1999; Gibbons and Machin, 2003; Bayer et al., 2007; Fack and Grenet, 2010), while recent Beijing work studies price, sorting, and spatial responses to multi-school zoning (Guan et al., 2024; Yang et al., 2024; Dai et al., 2025, 2026; Tong et al., 2024; Chen et al., 2026). Existing school-market papers already reach beyond capitalization: Shao et al. (2023) document timing and transaction-volume responses around a later Xicheng reform; Zheng et al. (2025) show that suppressing school-quality labels changes buyer visits, final listing prices, transaction prices, and time on market; and Ding and Itoh (2023) study a property-level school-admission reuse restriction in Shanghai. The object here is narrower: a sale-triggered change in the bundle being exchanged, studied across distinct for-sale observation universes.¹

Second, the paper extends evidence that nonportable incumbent benefits suppress mobility—including property-tax grandfathering and mortgage-rate lock-in (Ferreira, 2010; Fonseca and Liu, 2024; Batzer et al., 2026)—to a public right bundled with housing. The closest institutional analogue is Gonçalves et al. (2022): a Lisbon short-term-rental licence survived for incumbents but expired when the licensed home sold; they study registrations, dwellings sold, transaction prices, and Airbnb listings and prices. Related housing restrictions can also produce a large price–volume divergence (Somerville et al., 2020) or

¹Zhang (2022) describes a completed Beijing school-allocation project keyed to transaction timing, but no public manuscript or findings were located as of August 2026.

move resale listings, days on market, and sale-to-list ratios (Qiu and Wang, 2026). I bring these ingredients together for a title-vintage public-school right by separating completed resales, active for-sale stock and asking prices, and prices and marketing time conditional on completion. Third, the model treats the attachment rule itself as a policy choice and shows that reliance protection and protection conditional on non-sale are conceptually distinct.

The evidence supports a deliberately bounded conclusion. The headline estimate is a benchmark-sensitive, short-run relative exchange gap, not an absolute market collapse. The distinct observation universes do not identify full-market entry, withdrawal, or a sale hazard, and the data do not identify a structural entitlement value or welfare ranking. The completed-flow point estimates are much larger than the available selected-price points, but those objects come from different risk sets and do not identify a causal quantity–price decomposition.

The remainder of the paper presents the institution, data, and design; establishes the completed-flow fact; develops the model; tests its implications across the exchange funnel and longer horizons; and compares transition rules.

2 The Policy: Grandfathering Ends at Sale

2.1 School access as a housing entitlement

Public primary-school access in Beijing has historically been tied closely to household registration and residence. In the single-school regime, an eligible address was mapped to one responsibility school. That mapping gave ownership an entitlement-like component: a household buying a home in the zone could use the address when applying for admission, subject to household-registration and occupancy requirements. A large empirical literature finds that school access capitalizes into housing values because residence and enrollment are jointly chosen (Black, 1999; Bayer et al., 2007; Fack and Grenet, 2010). In Beijing the institutional link is particularly visible in the category of “school-district housing.”

The relevant treatment is not simply whether a home was located in Haidian. Every community in the estimation sample was in the district, and every one faced the reform. Former key-school status proxies for the value of the prior single-school mapping that was lost; it is not an observed household-specific treatment intensity. I classify schools using the 67-school union reported in Appendix Table A1 of Zheng et al. (2025), which combines a 2013 China Study Center ranking with a widely circulated online list; it is not an official government designation. A community enters the treated group only when a visibly dated 2017 enrollment-notice image names it in the school’s admission scope. Campus-brand, renamed-school, and loose affiliation links are excluded. Ordinary controls come from the pre-reform platform school map, exclude every strict or affiliated key-school match, and are chosen by geography within Haidian.

2.2 The title-vintage rule and its implementation

The policy document was dated April 28, 2018 and published April 30. In the author’s translation, its operative clause states: “Beginning January 1, 2019, houses newly registered and issued a real-property certificate will no longer correspond to one school; enrollment will follow multi-school zoning” ([Haidian District Education Commission, 2018a](#)). A December 2018 district notice repeated the same effective date and rule ([Haidian District Education Commission, 2018b](#)). The 2018 notice does not publish each address’s post-reform school set, assignment probability, or capacity constraint. Hence the policy created a known title-vintage cutoff but not an observed continuous treatment dose.

An official district retrospective reports that computerized allocation for families who bought after the title cutoff was first used in 2021 ([Haidian District, 2022](#)). The January 2019 design therefore identifies market responses to the title rule and its expected assignment consequences, not responses to contemporaneous realized lottery draws.

Later implementation materials make the post-transfer uncertainty more concrete without revealing a transaction-specific treatment dose. For the first post-cutoff cohort allocated by computer in 2021, the registration school was the first choice and guardians selected the remaining options. Applicants were admitted directly when demand did not exceed the allocation plan and were allocated by computer when it did. The number of options displayed depended on remaining places in the local area and could differ across families. Post-transfer access was therefore capacity-contingent, but the public materials do not report school-level plans, applicant counts, family-specific menus, or assignment probabilities for the transactions in the estimation window ([Haidian District Education Commission, 2021](#); [Beijing Evening News, 2021](#)).

The title-vintage clause is economically central. A pre-cutoff title remained under the prior single-school rule while held; a buyer whose acquisition generated a post-cutoff title entered the new assignment regime. Admission still depended on household-registration and occupancy requirements. The reform therefore did not simply lower a neighborhood amenity for everyone at once; it made the governing school-assignment rule conditional on whether the asset was transferred.

Figure 2 makes the economic contrast explicit. If the incumbent keeps the dwelling, the physical house and the prior single-school mapping remain bundled. If the dwelling is sold, the physical asset passes to the buyer but the assignment regime resets. Price can divide the surplus from the house; it cannot make the buyer’s post-transfer mapping identical to the incumbent’s retained mapping. The empirical exposure measure asks where that reset was likely to matter most.

The economic prediction is ambiguous outside a narrow model. Before the cutoff, households could accelerate eligible transfers to preserve the prior single-school mapping. After the cutoff, the expected school bundle could fall, assignment uncertainty could rise, and buyers and sellers could take longer to locate an acceptable surplus. A broad grace-period surge is therefore one possible response, but not a logical necessity. The analysis treats that

Figure 2. School-assignment rules before and after a title transfer



Notes: Keeping a pre-cutoff title preserves the prior single-school rule. Selling generates a new title, so the buyer enters multi-school assignment. The physical dwelling transfers; the incumbent’s assignment rule does not. Admission remains subject to household-registration and occupancy requirements.

Figure 3. Policy and identification timeline



Notes: The shaded regions are the sixteen-week estimation windows around January 1 in 2018 and 2019. January 1, 2018 supplies the identical-calendar seasonal comparison and is not a policy cutoff.

announcement-period response and the local enforcement response as distinct predictions; both were specified before reconstruction of the missing completion dates.

The cutoff is legally defined by registration and certification, not by the platform’s transaction-completion timestamp. Administrative processing may separate the two. On March 1, 2019, Haidian issued its first certificate under Beijing’s newly introduced integrated transaction–tax–title-registration service; the official account states that total administrative handling was compressed to four working days ([Beijing Municipal Commission of Planning and Natural Resources, 2019](#)). The source does not establish that the platform timestamp is the certificate-issuance date. I therefore treat this concurrent change as a timing threat, not as evidence that the school rule caused the broader winter-to-spring rise. A districtwide process change can generate the triple difference only if its take-up or transmission into the platform clock differs systematically by former school status. These timing distinctions are why I use sixteen-week windows, show bandwidth and donut sensitivity, and avoid the language of unit-level sharp treatment assignment. The estimand is a differential completed-flow response around enforcement, not a local average treatment effect for homes one day apart in deed issuance.

3 Data: Three Views of Housing Exchange

The observed outcome is housing exchange, not household migration: the archive does not link sellers or buyers across addresses.

The project combines three empirically distinct objects. First, the canonical platform archive contains completed resale transactions and supplies the main outcome. Within that same archive, I use formerly key, matched ordinary, other Haidian, and Beijing-outside-Haidian observations as four mutually exclusive descriptive benchmark groups for the observed policy-year winter-to-spring rise; they are neither independent research designs nor untreated policy controls. Second, posting and completion clocks within the archive construct a selected pipeline consisting only of listings observed to complete by the archive end. Third, independent public crawler snapshots observe active offers at particular dates. Keeping these objects separate is not semantic housekeeping: it determines which denominators exist and which market stages can be studied.

3.1 Completions, successful-listing spells, and active stock

Completed transactions. The main archive records successful exits from the resale process. A month-only source completion date is still a completed transaction when the row carries a positive deal price and a marketing-duration clock. Missing a parsed day is therefore not evidence of non-sale. These records identify the timing and location of platform-recorded completions, but not the number of properties ever offered.

The observed-completer pipeline. For completed records, posting dates reveal when each successful listing entered the platform pipeline. This creates exact posting, stock, and exit accounting within the selected set $I_i = 1$ of listings recorded as complete before the archive ends in January 2021. Both January–April posting cohorts used below receive at least 540 days of common follow-up, but conditioning on $I_i = 1$ still excludes withdrawals, unresolved listings, and later completions by construction.

Public active-stock snapshots. Four independently collected public files contain active listings: a November 29, 2017 crawl with exact listing IDs (Huang, 2017), a late-March 2018 cross-section distributed with the H4M data (Zhao et al., 2022), an archived project crawl collected community-by-community on February 27, 2019, and a public November 8, 2019 Lianjia community ledger.² The 2017 and 2019 ledgers use the same Lianjia community identifier and active-count field across different crawler builds. None supplies a request-success ledger. The files do not form a property-level transition panel, and their union does not reveal entry, withdrawal, relisting, or terminal states. I use only exact-name, stable-ID community links and make the cross-crawler comparability assumption explicit.³

²The November-2019 file is `lianjia-scrawler/community_info.csv` in [UltimateJupiter/lianjia](#), commit `c864b8c`. An earlier full export in the same commit records identical active-listing and housing-unit fields for all 54 linked communities 1.5–3 hours earlier.

³The provider’s auxiliary listing/attention export also includes rows without published transaction prices.

Asking-price moments require a positive valid offer in both waves. The November–February comparison therefore contains 50 communities and 958 valid listing-level observations, while the March–February comparison contains 36 communities and 836 observations. These supports must not be used to estimate aggregate active stock because conditioning on positive listings discards source-native zero-stock community–waves. The preferred stock comparisons retain those zeros on broader 54- and 37-community supports. The exact-linked November-2017–November-2019 ledger comparison also contains 54 communities; its 2019 field labelled total housing units supplies the scale used in Figure 1. That field is post-treatment Lianjia metadata rather than an official dwelling register.

3.2 Transaction archive and estimation sample

The transaction archive is a processed export from a proprietary listing platform covering Beijing resale records posted from 2010 through January 2021; the analysis period is 2015–2020. The canonical input contains 586,312 completed transactions after the project’s cleaning screens and supplies both the four geographic archive benchmarks and the estimation frame. Every retained record has a positive transaction price, posting date, and marketing duration. Some observations lack a parsed day-level completion date because the source date is month-only, but that field does not indicate that the property failed to sell. The analysis therefore uses counts of platform-recorded completed transactions rather than a listing-level sale probability.

For the preferred notice-verified Haidian frame, the archive supplies 7,340 resolved transactions across fifty-nine communities: nineteen exposed communities and forty ordinary-school comparisons. Of these, 1,173 fall inside the four sixteen-week estimation windows. The regression contains 236 community–experiment–side observations. The online appendix summarizes the estimation samples and reports the archive-wide measurement audit.

3.3 Completion-date measurement

Let a_i be listing i ’s posting date and W_i its recorded marketing duration. When an exact completion date d_i is present, I retain it. When it is missing, I use the fixed interval

$$d_i \in [a_i + W_i - 1, a_i + W_i]. \quad (1)$$

The lower endpoint follows the modal inclusive-day convention; the upper endpoint is the one-day sensitivity. The convention was fixed before re-estimating the policy coefficient.

The validation sample contains 414,979 transactions with both an exact date and the duration clock. Reconstructed and exact duration have correlation 0.99999, and all but

In the fifty-nine-community frame, only nineteen such rows have 2018 posting years and sixty-one have 2019 posting years; their observed status dates begin on April 23, 2019 and are asynchronous. They confirm that non-completer records exist but do not form a January 2019 risk set.

one reconstructed date lies within one day of the reported date. In the broader sixty-one-community matched frame, 2,515 of 7,410 dates use the reconstruction; twenty-seven of those reconstructed dates (1.1 percent, or 0.36 percent of all rows) change cutoff-cell membership between endpoints. Both endpoints remain visible throughout the completed-flow analysis.

A stricter audit re-dates every row from posting date and duration, ignoring whether an exact day was recorded. The raw, designated, and all-pre signs remain negative; designated PPML ratios are 0.430 and 0.454, almost identical to 0.422 and 0.462 under the mixed clock. Thus clock mixing does not mechanically generate the result. The harder source-format threat remains: exact-day observability itself has a DDD of $-0.316/-0.358$, and reconstructed-only rows do not reproduce the negative flow estimate. Both formats are positive completed transactions, but the archive does not document platform market share, off-platform trades, or a stable coverage denominator. Treatment-neutral relative platform capture is therefore a maintained condition, not a verified fact.

3.4 Historical exposure and comparison communities

Historical exposure is fixed from 2017 school-admission notices. The exposed sample contains nineteen communities tied to seven schools externally classified as key. An outcome-blind archival reconstruction of the 2018 notice images verifies that all nineteen exposed communities remained assigned to the same seven schools one year later; this check neither selects units nor rematches the design. The images are retained with source pages and hashes, but come from a public secondary archive rather than an authenticated district database. Among the original ordinary-school comparisons, thirty-one communities appear literally in the 2017 notice text and nine are linked by an exact, unambiguous property or address alias. The preferred sample retains those forty controls; an exact-notice sensitivity retains only the thirty-one literal matches. Neither restriction rematches the treated communities.

Each exposed community is matched to as many as five nearby eligible ordinary-school communities, prioritizing the same pre-reform school district and otherwise remaining within Haidian. Reused controls receive inverse-reuse weights, normalized within the matched design. Inference clusters by historically assigned school. The preferred sample contains twenty-one school clusters, seven of which are exposed, so the headline analysis exhausts all 2^{21} assigned-school Rademacher sign vectors under the restricted-null, fully studentized wild procedure and also reports leverage-aware CR2/Satterthwaite inference ([Pustejovsky and Tipton, 2018](#)) together with leave-one-out diagnostics. This enumeration removes Monte Carlo error but is not assignment-based randomization inference.

The external key-school classification is not created from the policy outcome. As an ex ante validation, I estimate a school component from 2015–2016 transaction prices and ask whether it predicts 2017 values before the announcement. The component lowers 2017 transaction-level mean squared error, its school-mean validation slope is 0.934, and formerly key schools score 0.851 standard deviations above ordinary schools. The online appendix

reports the full diagnostic and explains why the externally classified binary ordering remains the primary exposure measure.

3.5 Outcome construction and aggregation

The main dependent variable is the number of platform-recorded completed transactions in each community–experiment–side cell. Each community appears four times: before and after January 1 in the 2018 seasonal experiment and before and after January 1 in the 2019 policy experiment. Zero-count cells are retained. The preferred linear specification applies the inverse hyperbolic sine to accommodate zeros without adding an arbitrary constant; PPML estimates the same saturated contrast on the count scale.

The sixteen-week window was fixed to accommodate the separation between posting, platform completion, and title certification while remaining local to enforcement. I report the complete bandwidth and donut family rather than choosing a window after reconstruction. The average gap becomes more negative as the window expands, but no individual four-week bin survives family-wise adjustment. The design therefore identifies a missing exposed-group New Year rebound over sixteen weeks, not a discontinuity on January 1 or the mechanism generating its timing. Longer calendar-year and two-year outcomes are separate estimands.

For the timing analysis, marketing duration is the interval from posting to platform completion. A separate attention file is linked to the completed archive only through unique, outcome-blind exact matches. The linked sample contains 855 archive-observed sales for physical showings and duration and 755 for page views. All attention outcomes are secondary because they condition on observed completion; cumulative outcomes also mechanically load on exposure time.

4 Empirical Strategy: A Seasonal Triple Difference

4.1 Estimand and comparison groups

Index communities by c , cutoff experiments by $y \in \{2018, 2019\}$, and sides of January 1 by $s \in \{\text{before}, \text{after}\}$. Each side contains sixteen weeks. Let G_c indicate a formerly key-school community, R_y the real 2019 cutoff experiment, and P_s the post-January 1 side. For outcome Y_{cys} , the estimating equation is

$$\text{asinh}(Y_{cys}) = \alpha_c + \lambda_{ys} + \rho(G_c \times R_y) + \phi(G_c \times P_s) + \tau_F(G_c \times R_y \times P_s) + \varepsilon_{cys}. \quad (2)$$

Community fixed effects absorb permanent level differences, and λ_{ys} absorbs every experiment-by-side shock common to exposed and control communities. The lower-order exposure interactions allow the groups to differ across experiments and in average seasonality. The coefficient is

$$\tau_F = [\Delta_{A-B}(G = 1) - \Delta_{A-B}(G = 0)]_{2019} - [\Delta_{A-B}(G = 1) - \Delta_{A-B}(G = 0)]_{2018}, \quad (3)$$

where Δ_{A-B} is the after-minus-before change. A negative τ_F means that the exposed-control seasonal difference moved downward more around the 2019 cutoff than around the identical 2018 calendar date; it does not by itself imply that exposed counts fell in absolute terms. I also estimate the count model by PPML with the same design, weights, and fixed effects. Its reported relative post ratio is $\exp(\hat{\tau}_F^{\text{PPML}})$.

4.2 Identifying assumption and inference

The causal interpretation requires two restrictions. First is parallel differential seasonality: absent the reform, the exposed-control difference in the sixteen-week winter-to-spring change in 2019 would follow the same-calendar differential in the chosen pre-policy benchmark. This is more credible than a raw before-after comparison because housing activity is highly seasonal (Ngai and Tenreyro, 2014), but the choice of benchmark matters. Second is treatment-neutral relative platform capture: absent a true flow response, the archive’s treatment-relative probability of recording a completed transaction would not change across the same cells. The homogeneous-clock audit addresses timing construction, not this coverage condition. The successively broader archive benchmarks establish that completed exchange rose broadly in the policy-year window, but they neither choose the counterfactual nor verify platform capture.

Five threats remain. First, another January 2019 shock could differentially affect precisely the seven key schools. Second, the processed archive could undergo a treatment-correlated coverage change. Third, the historical school map could be wrong. Fourth, ordinary-school communities could receive spillovers from the same districtwide reform. Fifth, the platform completion date may misalign with administrative certification, including through the concurrent integrated-registration change described above. The design addresses, but cannot eliminate, these concerns through a prior-year calendar experiment, rolling false cutoffs, notice verification, both timing endpoints, local geographic controls, school-cluster inference, leave-one-community and leave-one-school estimates, and bandwidth/donut windows. For the fourth threat, Section 5.2 finds no detectable larger completed-flow increase among control communities nearest the exposed group, while leaving diffuse or delayed reallocation unresolved.

The control group is not untreated in the policy sense. Both groups entered multi-school zoning. The estimand contrasts replacement of a prior single-school mapping attached to a key school with replacement of an ordinary-school mapping. The DDD is a relative-exposure contrast that can combine exposed underperformance with any reform-induced gain or displaced activity in ordinary communities. It is not the district-average effect. Spillovers that move ordinary controls in the same direction attenuate the differential; spillovers of the opposite sign can amplify it.

Two timing predictions were fixed before reconstruction of missing completion dates: announcement-period acceleration and the sixteen-week cutoff comparison. The former is unsupported; the latter has the predicted sign under both completion-date endpoints

and notice restrictions. The online appendix reports the complete timestamped protocol, including specifications that do not meet the original thresholds. These protocols were internal rather than externally preregistered.

4.3 Outcome-blind spatial validation

An outcome-blind spatial design replaces the original many-to-one frame with a global maximum-cardinality, minimum-total-distance assignment that never uses transaction outcomes. It pairs all nineteen exposed communities without replacement to distinct notice-verified controls within 2.384 kilometers and gives each of the seven exposed schools equal weight. This is a local-counterfactual validation, not a boundary discontinuity or a second source of policy variation: the data contain community points rather than historical catchment polygons, and nearby controls may receive displaced transactions. The online appendix reports the pair map, construction, calipers, road-network and geographic-surface checks, Lunar-New-Year exclusions, and pre-policy diagnostics ([Butts, 2023](#)).

5 Completed Exchange at the Title Reset

The headline result is a short-run divergence in completed flow. This section documents that gap and asks how much it depends on the seasonal benchmark. The broad announcement-to-cutoff acceleration prediction is unsupported at either completion-date endpoint. Estimates become more negative as the symmetric window expands, but no individual four-week bin survives family-wise adjustment. I therefore interpret the outcome as a missing exposed-group New Year rebound over sixteen weeks, not a one-day discontinuity or evidence on why the timing unfolds.

5.1 Formerly key communities do not join the 2019 spring rise

The policy-year pattern is visible beyond the selected comparison group. Across the same sixteen-week windows, completed transactions rise in Beijing outside Haidian, in Haidian communities outside the fifty-nine-community frame, and most strongly in matched ordinary-school communities; formerly key-school communities remain approximately flat. The 2019 after-to-before ratios are 0.96–1.08 in formerly key communities, 1.67–1.79 in ordinary communities, 1.20–1.33 in other Haidian, and 1.29–1.47 outside Haidian. These mutually exclusive groups are descriptive benchmarks from the same archive, not independent replications, assignment-verified controls, or untreated estimates. They document the observed rise against which the exposed segment’s flat path is economically salient.

The full 59-community count sample gives the same anatomy before regression adjustment. In 2019, design-weighted means move from 6.37 to 6.11 and from 6.00 to 6.47 in formerly key-school communities under the two clocks, compared with 2.91 to 4.87 and

2.79 to 4.99 in ordinary-school communities. The exposed counts therefore do not show a large observed contraction. Their higher levels partly reflect larger communities; the exact-linked normalization in Figure 1 puts both groups near three completions per 1,000 Lianjia-recorded units before the cutoff. In the same 2018 calendar comparison, the exposed group rises unusually strongly. The seasonal regression subtracts that difference.

Table 1 reports the designated completed-flow DDD. Relative to 2018 differential seasonality, the asinh coefficients are -0.918 and -0.762 , and separate PPML estimates imply ratios of 0.422 and 0.462 . Under that one-year benchmark, these ratios represent exposed-relative gaps of 54–58 percent; they do not imply that exposed counts or total Haidian transactions fell by those percentages. Because 2018 is unusually favorable to the exposed group, these are the high-end magnitudes rather than a benchmark-free effect.

Exhaustively enumerated restricted-null school-wild tests for the asinh coefficients yield p -values of 0.012 and 0.035 across the two date conventions. CR2/Satterthwaite inference is more conservative at the upper endpoint, where the p -value is 0.062 . Exploratory Holm adjustment across five benchmark definitions gives $.067$ and $.178$. I therefore emphasize the stable sign and the full benchmark range rather than a binary endpoint-specific significance claim. The PPML ratios provide scale; their conventional school-cluster tests are asymptotic. Literal notice-text controls, retained without rematching, imply similarly large relative gaps.

Table 1. Completed transaction flow around the January 2019 title-vintage cutoff

	Lower completion date		Upper completion date	
	Before	After	Before	After
<i>Panel A. Raw completed transactions per community</i>				
2018 identical-calendar comparison				
Formerly key-school communities	6.89	10.21	6.47	10.63
Ordinary-school comparison communities	3.16	3.44	2.92	3.68
2019 policy experiment				
Formerly key-school communities	6.37	6.11	6.00	6.47
Ordinary-school comparison communities	2.91	4.87	2.79	4.99
Observed 2019 change: formerly key-school (percent)		-4.1		7.9
Observed 2019 change: ordinary-school (percent)		67.1		79.2
<i>Panel B. Relative, seasonally netted regression estimates</i>				
Former key-school $\times$ After Jan. 1 $\times$ 2019		-0.918**		-0.762**
CR2 standard error		(0.296)		(0.345)
Restricted-null school-wild p -value		[0.012]		[0.035]
CR2/Satterthwaite p -value		[0.017]		[0.062]
PPML relative completed-flow ratio		0.422**		0.462**
School-clustered PPML p -value		[0.043]		[0.037]
Transactions (community-window cells)		1,173 (236)		1,173 (236)
<i>Panel C. Literal-notice assignment sensitivity</i>				
Exact-notice controls: asinh DDD		-0.870**		-0.715*
Exact-notice controls: school-wild p -value		[0.022]		[0.056]
Exact-notice controls: PPML relative-flow ratio		0.434*		0.483*

Notes: Each period contains sixteen weeks. Panel A reports inverse-reuse-weighted raw means for the 19 formerly key and 40 ordinary communities. Panel B reports the seasonally netted asinh DDD and a PPML ratio from the preferred specification with community and experiment-by-side fixed effects. Negative coefficients and ratios below one denote a relative shortfall, not a decline in formerly-key levels. Parentheses contain CR2 standard errors; brackets contain two-sided p -values. Stars use assigned-school tests: *** $p < .01$, ** $p < .05$, and * $p < .10$. Lower and upper are the endpoints of the one-day completion-date interval. Panel C retains literal notice-text controls without rematching; full specification and inference details appear in the online appendix.

5.2 What the seasonal counterfactual identifies

The most consequential diagnostic is the seasonal benchmark. Table 2 keeps the preferred 59-community frame fixed. The raw 2019 exposed-relative seasonal coefficients are $-0.523/-0.421$, with PPML ratios $0.574/0.602$. Subtracting 2016, 2017, their mean, or the full 2016–2018 mean preserves the negative sign but produces PPML ratios from 0.507 to 0.668. Subtracting the unusually positive 2018 comparison produces the largest gap, $0.422/0.462$. These multiyear comparisons were added after observing the adverse placebo and are diagnostic rather than pre-specified. The direction is considerably more stable than the magnitude.

Other diagnostics do not overturn the sign. Every community and assigned-school leave-out remains negative, all pre-specified bandwidth and donut specifications are negative, and the outcome-blind spatial match yields about -0.72 at either endpoint. Uniform posting-plus-duration clocks yield designated PPML ratios of $0.430/0.454$. The spatial estimate weakens at a 1.5 kilometer caliper and under quadratic geographic surfaces. These checks limit particular composition concerns but do not repair the differential-seasonality or platform-capture assumptions.

Exhaustive fixed-weight relabeling points the same way without relying exclusively on 2018. For the raw 2019 DD, the actual seven-school grouping ranks 2,427th and 2,589th from the most negative coefficient among all 116,280 labelings; one-sided coefficient tails are $.021/.022$ and two-sided studentized tails $.051/.047$. Under the all-pre mean, the corresponding tails weaken. The designated-2018 DDD ranks 276th and 1,743rd. Because key-school status was not assigned and the frame and weights were constructed around the actual map, these are finite label-sensitivity descriptions, not permutation, randomization, or design-based inference.

Table 2. The completed-flow gap across seasonal benchmarks

Seasonal comparison	Lower completion date			Upper completion date		
	Asinh	Wild p	PPML ratio	Asinh	Wild p	PPML ratio
Raw 2019 after–before difference	-0.523	.037	0.574	-0.421	.035	0.602
2019 minus 2016	-0.173	.714	0.637	-0.070	.876	0.668
2019 minus 2017	-0.509	.139	0.612	-0.474	.099	0.638
2019 minus 2018 (designated)	-0.918	.012	0.422	-0.762	.035	0.462
2019 minus mean of 2016–2017	-0.341	.333	0.597	-0.272	.432	0.623
2019 minus mean of 2016–2018	-0.533	.067	0.507	-0.435	.119	0.537

Notes: The first row is the formerly-key–ordinary difference in the 2019 after-minus-before change, with community and side fixed effects. The remaining rows subtract the named same-calendar seasonal comparison and include community and year-by-side fixed effects. Every estimate uses the frozen 59-community frame and inverse-reuse weights. Wild p -values exhaust or simulate the restricted-null assigned-school sign distribution for the asinh coefficient; PPML ratios use the same design but conventional clustered inference. The benchmark family was assembled after the adverse 2018-placebo diagnostic and is therefore sensitivity analysis. Holm-adjusting the five benchmarked asinh tests gives $.067/.178$ for the designated row.

6 A Match-Threshold Model of Trade, Timing, and Selected Prices

The evidence to be organized is a short-window relative-flow gap, a conditional active-stock–flow mismatch, and small, imprecise selected-price point estimates. The model asks when a right attached to non-sale becomes a wedge to trade. Its organizing object is the bilateral match threshold: a title reset can raise that threshold, remove marginal matches, and change selection into observed prices. The model does not estimate the wedge or identify the primitive behind the flow response. It shows how one mechanism can organize the observed configuration and how alternative attachment rules change the match set. The framework is deliberately partial equilibrium and uses standard housing-search and bargaining logic (Wheaton, 1990; Genesove and Han, 2012; Head et al., 2014; Han and Strange, 2015).

6.1 Sale raises the bilateral match threshold

Consider a homeowner deciding whether to retain a dwelling and a prospective buyer considering the same dwelling. Their reservation value and willingness to pay are

$$R_s = \bar{R}_s + a_s q_0, \quad (4)$$

$$V_b = \bar{V}_b + a_b \mu_1, \quad (5)$$

where q_0 is the value of the prior single-school mapping conditional on enrollment eligibility, $\mu_1 = \mathbb{E}[Q_1]$ is the expected value of the post-transfer assignment bundle, and $a_s, a_b \geq 0$ are household-specific school-use values. Moving costs, housing-match quality, and outside options are contained in $\bar{R}_s$ and $\bar{V}_b$. Define entitlement-free match surplus $X = \bar{V}_b - \bar{R}_s$ and the match threshold under title-contingent grandfathering

$$\Delta^T = a_s q_0 - a_b \mu_1. \quad (6)$$

The threshold has a useful decomposition:

$$\Delta^T = \underbrace{(a_s - a_b)q_0}_{\text{ordinary heterogeneity in use values}} + \underbrace{a_b(q_0 - \mu_1)}_{\text{buyer-side title-reset component}}. \quad (7)$$

The first term can arise even when an amenity is fully transferable. The second arises because the assignment bundled with the property changes when title transfers. It is this second component that distinguishes the policy from an ordinary capitalization problem. A bilateral match has nonnegative surplus if and only if

$$X \geq \Delta^T. \quad (8)$$

The superscript T denotes title-contingent grandfathering. The same physical house can therefore be mutually acceptable when the school right transfers and unacceptable when

a sale resets eligibility. By contrast, an amenity change that shifts R_s and V_b by the same amount is capitalized into the transaction-price level while leaving match surplus, and hence equation (8), unchanged. That distinction separates the paper's transaction wedge from a standard capitalization exercise.

Conditional on predetermined property and market states, let X have cumulative distribution F , density f , and survivor function $\bar{F} = 1 - F$.

Proposition 1 (A higher threshold removes marginal matches). *The probability that a potential match has nonnegative surplus is*

$$p(\Delta) = \mathbb{P}(X \geq \Delta) = \bar{F}(\Delta). \quad (9)$$

At an interior point with $f(\Delta) > 0$, $p'(\Delta) = -f(\Delta) < 0$. An increase from Δ_0 to Δ_1 removes the set $\{X : \Delta_0 \leq X < \Delta_1\}$ from the positive-surplus region.

Proof. Differentiate $1 - F(\Delta)$ with respect to Δ . The discrete change is $F(\Delta_1) - F(\Delta_0)$. $\square$

The model delivers a cross-sectional prediction only after an exposure ordering is supplied. If replacing a prior key-school mapping produces a larger expected loss than replacing an ordinary-school mapping, the post-transfer wedge rises more in the formerly key-school group. Completed activity in those communities can then fall behind ordinary-school communities. The empirical indicator supplies this ordering; it is not a household-specific estimate of Δ^T .

Remark 1 (Location composition during a common expansion). *Index mutually exclusive location groups by $g \in \mathcal{G}$. Let $M_g > 0$ denote the mass of potential bilateral match opportunities in a fixed window. Expected accepted flow and its location share are*

$$C_g = M_g \bar{F}_g(\Delta_g), \quad s_g = \frac{C_g}{\sum_{j \in \mathcal{G}} C_j}. \quad (10)$$

For the formerly-key group K , holding opportunity masses, surplus distributions, and other groups' thresholds fixed,

$$\frac{\partial s_K}{\partial \Delta_K} = -s_K(1 - s_K)h_{X,K}(\Delta_K) < 0. \quad (11)$$

Now consider a common expansion that scales every group's opportunity mass by $\rho > 1$. If the formerly-key threshold rises from Δ_{K0} to Δ_{K1} while other thresholds remain fixed, then

$$\frac{C_{K1}}{C_{K0}} = \rho \frac{\bar{F}_K(\Delta_{K1})}{\bar{F}_K(\Delta_{K0})} < \rho = \frac{C_{g1}}{C_{g0}}, \quad g \neq K. \quad (12)$$

Accepted flow in formerly-key locations can therefore remain flat while flow elsewhere rises if the acceptance loss offsets the common expansion. The fall in s_K concerns the location composition of accepted matches. It neither requires nor establishes that an individual buyer switches from a formerly-key to an ordinary community. If opportunity masses, seller entry, or growth rates respond differentially across locations, equations (10) and (12) remain a conditional benchmark rather than a structural decomposition. If comparison-group thresholds also change, the same relative conclusion requires the proportional decline in $\bar{F}_g(\Delta_g)$ to be larger in the formerly-key group; the exposure indicator alone does not establish that ordering structurally.

6.2 Completed-flow responses are signed; selected-price responses are not

Suppose successful matches divide surplus through Nash bargaining and the seller receives share $\theta \in (0, 1)$. The transaction price is

$$P = R_s + \theta(V_b - R_s) = R_s + \theta(X - \Delta). \quad (13)$$

Prices are observed only when $X \geq \Delta$. Let $m(\Delta) = \mathbb{E}[X \mid X \geq \Delta]$, $e(\Delta) = m(\Delta) - \Delta$ denote mean excess surplus, and $h_X(\Delta) = f(\Delta)/\bar{F}(\Delta)$ denote the hazard of X .

Proposition 2 (The trade response is signed; the selected-price response is not). *Holding R_s and θ fixed,*

$$\mathbb{E}[P \mid X \geq \Delta] = R_s + \theta e(\Delta), \quad (14)$$

$$\frac{\partial \mathbb{E}[P \mid X \geq \Delta]}{\partial \Delta} = \theta \{h_X(\Delta)e(\Delta) - 1\}. \quad (15)$$

Thus a larger wedge reduces the mass of acceptable matches but can lower, leave unchanged, or raise the mean price among completed trades.

Proof. Differentiating the conditional tail mean gives $m'(\Delta) = h_X(\Delta)[m(\Delta) - \Delta] = h_X(\Delta)e(\Delta)$. Because $e'(\Delta) = m'(\Delta) - 1$, substitution into equation (13) gives equation (15). $\square$

For an exponential surplus distribution, mean excess surplus is constant and the selected mean price does not move locally even though acceptable matches disappear. Other distributions can generate either sign. A completed-sale hedonic regression therefore cannot recover the transaction wedge by itself, and controlling completed flow for a post-policy transaction price would condition on an equilibrium outcome. This derivative is an organizing comparative static, not a structural interpretation of the empirical triple difference.

6.3 A higher threshold slows finite-horizon clearance

Suppose potential buyers arrive at Poisson rate λ and draw independent match surpluses from F . Conditional on a property being continuously offered, the clearance rate is

$$h_C(\Delta) = \lambda \bar{F}(\Delta), \quad (16)$$

and the probability of completion by horizon T is

$$H(T, \Delta) = 1 - \exp\{-\lambda \bar{F}(\Delta)T\}. \quad (17)$$

Proposition 3 (A wedge delays trade without requiring permanent missing trade). *For fixed λ and $T > 0$, a larger Δ lowers both the clearance rate and finite-horizon completion probability:*

$$\frac{\partial H(T, \Delta)}{\partial \Delta} = -\lambda T f(\Delta) \exp\{-\lambda \bar{F}(\Delta)T\} < 0. \quad (18)$$

Expected untruncated time to clearance is $1/h_C(\Delta)$ and rises with Δ . If two groups have strictly positive clearance rates, their cumulative-completion ratio converges to one as $T \rightarrow \infty$.

Proof. Differentiate equations (16) and (17). The limit follows because $H(T, \Delta) \rightarrow 1$ for every strictly positive $h_C(\Delta)$. $\square$

This benchmark illustrates how a short-window divergence could attenuate at longer horizons. The data do not identify continuous listing, clearance hazards, or property-level convergence: every duration observation conditions on eventual completion, and the separate two-year exercise has an adverse placebo. The duration and persistence estimates are therefore not empirical tests of this proposition. Withdrawals, seller entry, changing beliefs, spatial reallocation, and platform coverage can generate the same aggregate pattern.

To make that boundary explicit, let $L(\Delta)$ denote listing inflow and $W(\Delta)$ the withdrawal hazard. Active offered stock S evolves as

$$\dot{S} = L(\Delta) - [h_C(\Delta) + W(\Delta)]S, \quad S^*(\Delta) = \frac{L(\Delta)}{h_C(\Delta) + W(\Delta)}. \quad (19)$$

The sign of the stock response is unrestricted: a wedge can reduce seller entry and clearance simultaneously. A stock snapshot therefore cannot separate entry, acceptance, and withdrawal, while a completed-transaction archive contains exits but no full-market risk-set denominator.

6.4 One threshold organizes quantity, timing, and selected prices

Table 3 makes the fact-to-model mapping explicit. The model is asked to organize the joint configuration, not to convert each point estimate into a structural parameter.

An exponential benchmark nevertheless gives the flow ratio an illustrative scale. Under its memoryless survivor function, $\bar{F}(\Delta) = \exp(-\Delta/\sigma)$, mean excess surplus among successful matches is σ and the selected mean price is invariant to the threshold when R_s and θ are fixed. This is a convenient sufficient benchmark, not a distribution selected by an imprecisely estimated price null. If opportunity masses and relative platform capture are unchanged and PPML ratios are read as acceptance-rate ratios, the benchmark range 0.42–0.67 maps into a differential threshold increase of 0.40–0.87 mean excess-surplus units. Differential entry, arrival, coverage, or reallocation breaks that mapping. These are assumption-indexed normalizations, not estimates of a monetary entitlement value or welfare loss.

Seller reservation values, aggregate demand, seller entry, withdrawal, and congestion can produce similar flow–stock–price patterns (Stein, 1995; Genesove and Mayer, 2001; Anenberg and Ringo, 2024, 2026). The model therefore disciplines one economic interpretation rather than identifying a unique primitive. What is setting-specific is the legal source of the transfer-contingent asymmetry.

Table 3. Empirical facts and their model counterparts

Empirical fact	Model object	Discipline and remaining margin
Raw and benchmarked completed-flow ratios 0.57–0.61 and 0.42–0.67	Accepted matches, $\bar{F}(\Delta)$	A relative completion gap under differential seasonality and stable capture; opportunity masses remain unobserved
Flat formerly-key flow during a broad expansion	Accepted-location share s_K	A higher relative threshold can shift accepted flow; this is not buyer substitution
Late-March stock 0.923 versus flow 0.368–0.427	Offered stock and accepted flow	Conditional stage mismatch; crawler comparability and exposure alignment remain assumptions
Selected transaction-price coefficients $-0.011/-0.016$	Selected mean in equation (15)	Small, imprecise points among completers; the entitlement value remains unidentified

Notes: The table maps empirical objects to model counterparts. It does not treat the estimates as moments from a common risk set or use them to recover structural parameters.

7 Conditional Market-Stage Corroboration

The model motivates several market-stage objects, but the data observe them on different supports. Completed transactions define the headline conditional estimand. Public stock and asking-price snapshots provide only comparability-indexed corroboration, while posting clocks, duration, attention, and revisions are selected on eventual completion. I therefore do not combine them into a quantity–price decomposition or call any ratio a listing hazard.

Public active-stock snapshots provide the strongest conditional corroboration. Table 4 compares active stock and completed flow on identical community supports. On the late-March support, the active-stock ratio is 0.923 while completed-flow ratios are 0.368 and 0.427. Joint score tests of equal log responses give $p = .029/.033$; adjusting across the March and November comparisons gives a Holm value of .066. November produces the same point ordering—stock 0.830 versus flow 0.420/0.464—but weaker equality inference. The conclusion is conditional: comparable treatment-relative crawler coverage and alignment of a point snapshot with period exposure are not verified.

The additional exact-ID ledger comparison is deliberately weaker. It follows 18 of 19 exposed and 36 of 40 ordinary communities from November 29, 2017 to November 8, 2019. Lianjia-visible active listings more than double in both arms, with no detectable differential change (asinh interaction 0.008, school-wild $p = .971$; PPML ratio 1.059). This is not evidence that inventory is constant or equivalent across groups. The ledgers come from different crawler builds without request-success logs, and their housing-unit field is post-treatment platform metadata. The exercise shows that the flow sign survives exact-ID support, scale normalization, and every exposed-school deletion; it does not identify physical turnover or title exhaustion.

Table 4. Completed flow and public active stock on identical supports

Common support	Object	Relative ratio	95% interval	Communities
Nov. 2017–Feb. 2019 common support	Completed flow (lower date)	0.420	[0.167, 1.057]	54
Nov. 2017–Feb. 2019 common support	Completed flow (upper date)	0.464	[0.209, 1.033]	54
Nov. 2017–Feb. 2019 common support	Public active-stock contrast	0.830	[0.477, 1.447]	54
Late-Mar. 2018–Feb. 2019 H4 support	Completed flow (lower date)	0.368	[0.117, 1.154]	37
Late-Mar. 2018–Feb. 2019 H4 support	Completed flow (upper date)	0.427	[0.160, 1.142]	37
Late-Mar. 2018–Feb. 2019 H4 support	Public active-stock contrast	0.923	[0.557, 1.531]	37

Notes: Every stock-flow comparison uses identical community support within the named support. Completed-flow ratios compare the formerly-key/ordinary after-before ratio in 2019 with its identical-calendar 2018 analogue. Public active-stock rows instead compare exposed-relative point stocks between the two named crawler waves; they are not transaction-DDD estimands. Ratios below one therefore denote a lower exposed-relative completed-flow response or a lower cross-wave exposed-relative stock contrast, respectively; they do not imply a decline in exposed levels. Intervals use equation-specific school-clustered t critical values and are not joint confidence sets for the stock-flow gap. The public waves are separate crawler snapshots rather than a listing-transition panel. On the 54-community November support, the stock/flow point-ratio gaps are 1.97 and 1.79; joint school-score equality tests give $p = .085$ and $p = .137$ using the geometric mean of the equation-specific CR1 finite-sample corrections. The smaller H4 support gives larger point gaps but only four exposed-school clusters. This is cross-crawler comparability sensitivity, not an identified execution hazard.

Available price evidence is similarly bounded. On positive-listing common supports, community-mean asking-price contrasts are -0.003 and -0.008 , with 95 percent intervals $[-0.051, 0.045]$ and $[-0.062, 0.046]$. Medians move -0.002 and $+0.003$, while listing-level hedonic points are -0.012 and -0.015 . Among completed sales, transaction-price points are $-0.011/-0.016$. None is an equivalence result; the intervals permit moderate price movements, and no public-listing outcome survives its within-family adjustment. The evidence therefore establishes coexistence of a large relative-flow estimate with small and imprecise observed-price points, not that prices did not change.

The remaining mechanism searches do not survive. No selected-cohort clearance horizon survives family-wise adjustment, and the prior-year diagnostic reverses. Duration, attention rates, price revisions, completed-sale composition, unit-type interactions, dose, and diversion yield no endpoint-conservative survivor. Posting clocks exactly account for stock and exit only within eventual sales and therefore cannot identify offered-market entry. These tests provide no affirmative evidence that affected homes simply wait longer for better matches; they do not exclude that possibility. The data remain consistent with several alternatives: differential seller entry, demand, expectations, amenities, spatial reallocation, bargaining, or platform coverage.

8 Limits: Persistence and Diversion

The short-window result is not a permanent market-freeze estimate. A separately locked exercise yields an exposed-relative PPML ratio of 0.84 for calendar 2019 and 1.01–1.02 for 2020. Cumulative 2019–2020 flow relative to 2016–2017 gives 0.945 at both endpoints,

with wild $p = .815/.835$. A similarly sized placebo prevents interpreting this null as evidence of delayed completion or property-level convergence.

Nor do the data identify where the missing relative flow went. Pre-specified proximity tests do not show a larger increase among controls nearest the exposed group. Within nineteen outcome-blind spatial pairs, the formerly-key share falls by an imprecise 10–12 percentage points while combined pair flow shows no detectable differential. The pairs are not closed markets, and neither exercise establishes diversion. Diffuse reallocation, delayed transactions, suppressed exchange, and platform coverage remain observationally combined; the online appendix reports the full failed families.

9 Transition Design

The policy tension is not grandfathering itself but protection conditioned on non-sale. Under title-contingent grandfathering, retaining the asset preserves the old rule and transfer extinguishes it. The model separates that reliance objective from the attachment rule: protection can instead follow the property, the household, or a common transition date.

9.1 Attachment rules and the exchange margin

Under title-contingent grandfathering, the wedge is Δ^T in equation (6). If the old assignment follows the property after sale, if the incumbent household can carry it after moving, or if old and new owners face a uniform post-transition rule, respectively, the corresponding wedges are

$$\Delta^P = a_s q_0 - a_b q_0, \quad (20)$$

$$\Delta^H = -a_b \mu_1, \quad (21)$$

$$\Delta^U = (a_s - a_b) \mu_1. \quad (22)$$

Property portability raises what the buyer obtains from the dwelling. Household portability removes the incumbent's lost entitlement from the reservation value of selling. A uniform transition removes the title-vintage difference, but changes the incumbent's protected bundle even without a sale.

Proposition 4 (Portability weakly expands the match set). *Suppose $q_0 \geq \mu_1$ and $a_s, a_b \geq 0$. Holding school capacity, assignment probabilities, and outside housing options fixed,*

$$\Delta^T \geq \Delta^P, \quad \Delta^T \geq \Delta^H, \quad \Delta^T \geq \Delta^U. \quad (23)$$

Hence each portable or uniform rule weakly raises the bilateral match probability relative to title-contingent grandfathering. The ordering among P , H , and U is generally ambiguous.

Proof. The three differences are $\Delta^T - \Delta^P = a_b(q_0 - \mu_1)$, $\Delta^T - \Delta^H = a_s q_0$, and $\Delta^T - \Delta^U = a_s(q_0 - \mu_1)$, all weakly nonnegative. Apply Proposition 1. $\square$

Proposition 4 isolates a narrow transition-design margin. It is a partial-equilibrium match-rate ordering, not a welfare ranking. Its design implication is narrower: honoring reliance does not require making sale the forfeiture event. Table 5 shows exactly which side of the match each rule changes.

Table 5. What sale changes under alternative attachment rules

Rule	What happens at sale	Exchange-margin result	Main unmeasured trade-off
Title-contingent	Incumbent loses q_0 ; buyer receives μ_1	Benchmark threshold $\Delta^T = a_s q_0 - a_b \mu_1$	Sorting into legacy titles; capacity attached to old addresses
Property portability	Assignment q_0 follows the dwelling	Lowers threshold by $a_b(q_0 - \mu_1)$	Perpetuates address-based advantage and capitalization
Household portability	Incumbent carries q_0 ; buyer receives μ_1	Lowers threshold by $a_s q_0$	Travel, cross-neighborhood use, and destination capacity
Uniform transition	Incumbent and buyer both face μ_1	Lowers threshold by $a_s(q_0 - \mu_1)$	Weakens reliance protection; changes congestion and incidence

Notes: Threshold differences follow from Proposition 4 under $q_0 \geq \mu_1$ and $a_s, a_b \geq 0$. They order bilateral match probabilities in partial equilibrium, not welfare. A welfare comparison additionally requires preferences, assignment probabilities, capacity, realized school outcomes, and equilibrium sorting.

Which portable rule is appropriate depends on what the transition is meant to protect. Property portability preserves the address-based entitlement and lets a buyer obtain it through the house price, but can perpetuate the spatial scarcity rent. Household portability protects the incumbent family while allowing the dwelling to enter the new regime, but can create travel and destination-capacity costs. A common date-based transition restores symmetry when equal treatment is valued more than legacy protection. A time-limited household entitlement is one implementation worth evaluating, not a rule selected by the model or the data.

9.2 Scale, mechanism, and scope

An external transfer-tax yardstick. A sale-contingent entitlement reset enters bilateral surplus like a heterogeneous transaction cost, although it is not a statutory tax, raises no public revenue, and varies with household demand for schools. Transfer-tax studies show that modest sale-contingent wedges can move transaction volume and timing sharply (Dachis et al., 2012; Best and Kleven, 2018). These estimates provide an external mechanism benchmark, not a conversion of the Haidian coefficient into an implicit tax rate. California’s

household-portability provision for Proposition 13, which reduced measured lock-in among newly eligible movers, illustrates a different attachment rule in practice ([Ferreira, 2010](#)).

What the evidence identifies. The data identify a short-run formerly-key–ordinary difference in platform-recorded completed flow under parallel differential seasonality and treatment-neutral relative capture, not the primitive generating it. The title clause supplies a plausible transfer-contingent mechanism, and the conditional stock-flow ordering is consistent with an execution margin, but buyer arrival, seller entry, withdrawal, coverage, bargaining, and spatial reallocation remain combined. The result does not imply price rigidity: selection can move the mean price among completed trades in either direction. Welfare and a ranking of transition rules require assignment, preference, capacity, and migration data not observed here. Structural work integrating residential location and school choice illustrates the additional equilibrium objects such exercises require ([Bayer et al., 2007](#); [Cui et al., 2026](#)).

External validity rests on the mechanism rather than the local average effect: whenever ownership bundles a scarce public benefit and transfer extinguishes that benefit, sale can destroy bilateral surplus. The magnitude estimated here remains local to nineteen communities attached to seven formerly key schools in one Beijing district; it is not the average effect of multi-school zoning on Haidian or Beijing.

10 Conclusion

Beijing's title-vintage rule made sale the event that extinguished a legacy school-assignment mapping. On an exact-linked recorded-unit scale, formerly key and ordinary communities both begin near three platform completions per 1,000 units; ordinary communities then rise to about six while formerly key communities remain near three. The raw exposed-relative PPML ratio is 0.57–0.61. Same-calendar benchmarks imply 0.42–0.67, with the largest gap against an unusually strong 2018. The sign survives frozen frames, homogeneous completion clocks, count estimators, every school and community deletion, and recorded-unit normalization. It remains conditional on parallel differential seasonality and treatment-neutral relative platform capture; the exact-day-format audit makes the latter limitation visible.

A match-threshold model shows how a nonportable right can remove marginal trades without requiring a proportional change in the mean price among trades that complete. Public stock-flow and asking-price comparisons are consistent with that execution margin, but only conditional on crawler comparability; the data do not identify a listing hazard, delay, match quality, or a unique primitive. The model also separates reliance protection from protection conditioned on non-sale. Under its partial-equilibrium conditions, attaching protection to the household or property—or ending it on a common date—expands the feasible match set relative to a standing-title rule. This is not a welfare ranking, but it clarifies the policy choice: protecting reliance need not make exchange the event that destroys protection.

The paper has no external replication. A conservative building-ledger sensitivity makes Xicheng estimable, but its aggregate designated effect is null, its date-source components point in opposite directions, and older benchmarks reverse sign; Dongcheng's exploratory estimate is adverse and fails mapping and support gates. A decisive next test requires a verified address-to-title assignment registry rather than platform building metadata. Linking complete offered-market histories—entry, price revisions, withdrawal, relisting, and sale—to title registration and realized school assignment would then separate missing exchange from delay or withdrawal and permit a quantitative evaluation of alternative transition rules.

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Online Appendix

Grandfathering at Sale: Nonportable School Rights and Housing Exchange

Tai Lo Yeung

September 3, 2026

A Model Details

Let the incumbent's reservation value and a prospective buyer's willingness to pay be

$$R_s = \bar{R}_s + a_s q_0, \quad (\text{A1})$$

$$V_b = \bar{V}_b + a_b \mu_1, \quad (\text{A2})$$

where q_0 is the deterministic entitlement preserved by retaining the old title, μ_1 is the expected value of the post-transfer assignment bundle, and $a_s, a_b \geq 0$ are household-specific school-use values. Define entitlement-free match surplus $X = \bar{V}_b - \bar{R}_s$ and the transaction-contingent wedge

$$\Delta = a_s q_0 - a_b \mu_1. \quad (\text{A3})$$

Trade occurs if and only if $X \geq \Delta$.

Proposition A1 (A symmetric common shift leaves the model's trade set unchanged). *If a policy changes buyer value and seller reservation value by the same amount, it changes the bargaining price but leaves the trade set unchanged.*

Proof. Adding the same scalar d to R_s and V_b leaves $V_b - R_s$ and hence the positive-surplus indicator unchanged. $\square$

Holding the valuation loadings fixed and treating Δ as a scalar threshold, suppose X has cumulative distribution F and density f . The probability that a potential match trades is $p(\Delta) = 1 - F(\Delta)$.

Proposition A2 (A larger transaction-contingent wedge removes marginal matches). *At any interior point with $f(\Delta) > 0$,*

$$\frac{\partial p(\Delta)}{\partial \Delta} = -f(\Delta) < 0. \quad (\text{A4})$$

An increase from Δ_0 to Δ_1 removes the set of matches $\{X : \Delta_0 \leq X < \Delta_1\}$.

To characterize selection into observed prices, let the seller receive Nash weight θ and let $m(\Delta) = \mathbb{E}[X \mid X \geq \Delta]$. Conditional mean transaction price is

$$\mathbb{E}[P \mid X \geq \Delta] = R_s + \theta\{m(\Delta) - \Delta\}. \quad (\text{A5})$$

Writing $e(\Delta) = m(\Delta) - \Delta$ and $h(\Delta) = f(\Delta)/[1 - F(\Delta)]$, and holding R_s and θ fixed, gives

$$\frac{\partial \mathbb{E}[P \mid X \geq \Delta]}{\partial \Delta} = \theta\{h(\Delta)e(\Delta) - 1\}. \quad (\text{A6})$$

The selected-price response is unsigned even though match probability falls. If potential buyers arrive at rate λ , a listing that remains offered clears at rate $\lambda[1 - F(\Delta)]$. A larger wedge lowers clearance and lengthens the expected marketing spell for fixed λ , but duration among listings observed to complete remains selected and active stock also depends on listing entry and withdrawal.

B Measurement and Assignment Audit

The canonical transaction input is a processed proprietary listing-platform export containing 586,312 Beijing resale records posted from 2010 through January 2021. The policy analysis uses 2015–2020. All retained rows have positive transaction price, posting date, and marketing duration, so the archive contains completed transactions rather than an offered-listing denominator. The provider export does not document time-varying platform market share or off-platform transactions. The distributable package begins with the canonical analysis file and preserves every downstream policy transformation; reproducing the upstream provider-file cleaning step requires redistribution permission and a verified raw-to-canonical crosswalk.

A separate provider auxiliary listing-and-attention export does contain rows without published transaction prices. Within the fifty-nine-community frame, however, only nineteen such rows have 2018 posting years and sixty-one have 2019 posting years; their observed status dates begin on April 23, 2019 and are asynchronous across records. These rows establish that the raw bundle is not limited to completed transactions, but they do not recover the offered stock at the January 1 implementation date or form a listing-level policy risk set. The paper therefore does not reinterpret their missing transaction prices as observed failures to sell.

When the exact completion date is missing because the source date is month-only, the paper retains the fixed interval from posting date plus reported duration minus one day to posting date plus duration. In the 414,979-record validation sample that observes both clocks, all but one reconstructed date lies within one day of the reported date. In the broader matched frame, 27 of 2,515 reconstructed dates (1.1 percent, or 0.36 percent of all 7,410 transactions) change cutoff-cell assignment across endpoints.

Table A1: Transaction archive and completion-date measurement

Panel A. Archive-wide validation		
Record universe	Count	
Beijing completed-transaction archive	586,312	
Transactions observing exact and reconstructed clocks	414,979	
Reconstructed dates changing cutoff windows (broader frame)	27 of 2,515 (1.1%)	
Panel B. Retained-sample dates		
	Preferred assignment frame	Exact-notice sensitivity
Completed transactions in four estimation windows	1,173	1,029
Resolved archive support in retained communities	7,340	6,519
Archive records with reported exact date	4,863	4,303
Archive records with reconstructed date	2,477	2,216

Notes: The archive consists entirely of completed transactions. Reported exact dates are retained. Missing exact dates use the two fixed endpoints implied by posting date and marketing duration. The 414,979-record validation observes both clocks; all but one reconstructed date is within one day of the reported completion date. The broader-frame cutoff-window statistic has denominator 2,515 reconstructed dates (0.36 percent of all 7,410 records change windows).

The more consequential audit concerns source format rather than one-day timing. Exact-day observability

changes differentially across the four designated cells: its DDD is -0.316 and -0.358 (assigned-school-cluster $p = .017/.003$). Exact-day rows alone produce a larger negative flow estimate, whereas reconstructed month-only rows are near zero or positive. This composition shift means that treatment-neutral row capture is not verified.

To isolate mechanical clock mixing, I re-date every completed row from posting date and marketing duration and never use the exact-day flag. Table A2 shows that raw, designated, and all-pre estimates remain negative and numerically close to the mixed-clock results. Recorded exact dates differ from the uniform lower and upper assignments by at most one day. Thus source-format reclassification does not mechanically create the full-flow sign, but a treatment-correlated platform or upstream-format change could still proxy for unobserved coverage.

Table A2: Completion-clock and source-format audit

Panel	Specification	Lower date		Upper date	
		Asinh	PPML ratio	Asinh	PPML ratio
A. Core result using every positive completed-sale row					
	Raw 2019, mixed recorded/reconstructed clock	−0.523	0.574	−0.421	0.602
	Raw 2019, uniform posting-plus-duration clock	−0.522	0.577	−0.421	0.601
	2019 minus 2018, mixed clock	−0.918	0.422	−0.762	0.462
	2019 minus 2018, uniform clock	−0.894	0.430	−0.771	0.454
	2019 minus all-pre mean, mixed clock	−0.533	0.507	−0.435	0.537
	2019 minus all-pre mean, uniform clock	−0.528	0.508	−0.422	0.542
B. Source-format composition within the four designated cells					
	Exact-day observability DDD	−0.316	—	−0.358	—
	Exact-day rows only: flow DDD	−1.166	0.318	−1.166	0.318
	Month-only rows only: flow DDD	+0.028	0.850	+0.190	1.087

Notes: The uniform clock assigns every row to posting date plus recorded marketing duration, so it never uses the exact-day flag. Its lower and upper dates differ by one day. Recorded exact dates differ from those uniform endpoints by at most one day. Panel A therefore rules out mechanical mixing of exact and reconstructed completion clocks as the source of the sign. Panel B shows the unresolved threat: the probability that a completed row carries an exact day changes differentially across the four cells (cluster $p = .017/.003$), and month-only rows do not reproduce the decline. Both formats are positive completed-sale records, but a treatment-correlated platform or source-format change may still proxy for unobserved row coverage.

As a scale diagnostic, I also ask how much unrecorded flow would have to be added to the formerly-key 2019-after cell alone to set each PPML triple ratio to one, holding the other cells fixed. The break-even additions are 66–74 percent of recorded flow for the raw 2019 comparison, 50–68 percent for the conservative 2016, 2017, or 2016–2017 benchmarks, and 116–137 percent for the designated 2018 comparison. This is an adversarial single-cell sensitivity, not a bound: errors spread across several cells, treatment-related use of other platforms, or changes in the ordinary arm can erase the estimate under different patterns.

Figure A1 reports an unsmoothed calendar-month diagnostic under both endpoints of the fixed completion-date interval. It shows neither a broad announcement-to-cutoff acceleration nor a literal one-month break at January 1. The average exposed–ordinary gap becomes more negative as the window expands, but no individual four-week bin survives family-wise adjustment. I therefore interpret the evidence as a missing exposed-group New Year rebound over sixteen weeks, not a day-one discontinuity or an event-study path. Calendar-month assignment is much more sensitive to the one-day convention than the main sixteen-week cells: 232 reconstructed transactions move to an adjacent month between April 2018 and May 2019, whereas only 27 reconstructed transactions change membership in the main cutoff cells.

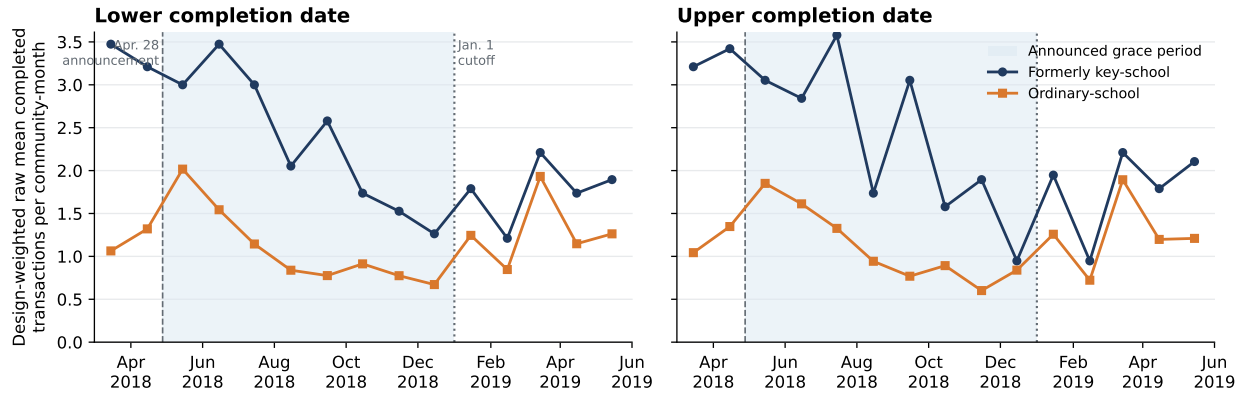


Figure A1: Raw monthly completed transactions under the two completion-date conventions

Notes: Each point is the inverse-reuse-weighted raw mean number of completed transactions per community-month in the preferred 59-community notice-verified frame. Zero-flow community-months are retained; no smoothing or significance filtering is applied. The two panels assign interval-dated transactions to the lower and upper endpoints of the fixed one-day completion interval. Raw group levels are not causal effects: both groups entered multi-school zoning, formerly key-school communities have higher baseline flow, and ordinary communities may receive reallocated transactions. April 2018 contains only three post-announcement days, and May 2019 lies beyond the formal sixteen-week post-cutoff window. Dates are platform-recorded completions rather than administrative title registrations.

The assignment frame uses visibly dated 2017 school-admission notices for every exposed community. The preferred comparison sample retains communities named literally in those notices or linked through an exact and unambiguous property or address alias. The exact-notice sensitivity retains only literal text matches. These restrictions preserve all nineteen exposed communities and do not rematch the design.

After freezing the design, I conducted an outcome-blind archival reconstruction of 2018 notice images to test whether the treated assignment was a transitory 2017 classification. Direct image review verifies the same assigned school in both years for all nineteen exposed communities and all seven exposed schools; Table A3 reports the school-level support. The 2018 images are preserved with page URLs and SHA-256 hashes, but are hosted by a public secondary archive rather than an authenticated district database. The main exposure therefore remains the frozen 2017 assignment, and the two-year check is used only as provenance and stability validation. Control-notice coverage is incomplete in 2018, so no control is added, deleted, or relabeled using that archive.

Table A3: Two-year historical-assignment validation for exposed communities

Historically assigned school	Exposed communities	2017 notice	Same assignment in 2018
Qiyi Primary School	3	Verified	3 of 3
Renda Affiliated Experimental Primary	1	Verified	1 of 1
Shangdi Experimental Primary School	3	Verified	3 of 3
CNU Affiliated Yuxin School	2	Verified	2 of 2
Yangfangdian Central Primary School	1	Verified	1 of 1
Zhongguancun No. 3 Primary School	1	Verified	1 of 1
Renda Affiliated Primary School	8	Verified	8 of 8
Total	19	19 of 19	19 of 19

Notes: The design is fixed from the 2017 notices. The 2018 column is a subsequent, outcome-blind stability audit based on direct review of dated notice images and does not rematch or select the estimation sample. Source pages, retained images, direct-read excerpts, and hashes are recorded in the replication archive. The images are hosted by a public secondary archive; the table therefore validates archival consistency, not district-database authentication.

Tables A4 and A5 provide stable row-level audit IDs and notice-image hash prefixes. The corresponding UTF-8 files in the replication folder preserve the original community and school names, matching phrase, public source URL, local image name, rationale, and full SHA-256 hash. Excluded candidates remain in the control audit rather than disappearing from the provenance trail.

Table A4: Exposed-community notice manifest

Audit ID	School ID	Longitude	Latitude	Notice page	SHA-256 prefix
HT01	3	116.290	39.967	11330	e3ce3d7c28
HT02	3	116.288	39.969	11330	e3ce3d7c28
HT03	3	116.289	39.965	11330	e3ce3d7c28
HT04	3	116.295	39.969	11330	e3ce3d7c28
HT05	3	116.292	39.972	11330	e3ce3d7c28
HT06	3	116.285	39.969	11330	e3ce3d7c28
HT07	3	116.292	39.969	11330	e3ce3d7c28
HT08	3	116.285	39.967	11330	e3ce3d7c28
HT09	4	116.298	39.977	11268	dadbcaffeb
HT10	30	116.358	40.071	12127	1927f52005
HT11	30	116.352	40.062	12127	1927f52005
HT12	46	116.338	39.914	11303	85eb12ddc1
HT13	59	116.320	40.039	11342	abf24ecb4b
HT14	59	116.323	40.041	11342	abf24ecb4b
HT15	59	116.318	40.039	11342	abf24ecb4b
HT16	61	116.305	39.895	11233	7e9d768cef
HT17	61	116.315	39.901	11233	7e9d768cef
HT18	61	116.313	39.902	11233	7e9d768cef
HT19	67	116.340	39.989	12175	d27f2ba414

Notes: Stable audit IDs map to the UTF-8 community and school names, complete notice URLs, dates, image filenames, and full SHA-256 hashes in `Replication/frozen_matched_community_frame_assignment_audited.csv`. All nineteen records are visibly dated 2017 notice-image links. School IDs are stable internal identifiers, not rankings.

Table A5: Ordinary-community assignment audit manifest

Audit ID	Evidence	Preferred	Literal	SHA-256 prefix
HC01	E	Yes	Yes	a3e6ae419b
HC02	U	No	No	5acd502cb8
HC03	E	Yes	Yes	5acd502cb8
HC04	E	Yes	Yes	5acd502cb8
HC05	A	Yes	No	5acd502cb8
HC06	E	Yes	Yes	5acd502cb8
HC07	E	Yes	Yes	5acd502cb8
HC08	E	Yes	Yes	5acd502cb8
HC09	E	Yes	Yes	5acd502cb8
HC10	E	Yes	Yes	b473d367d5
HC11	E	Yes	Yes	b473d367d5
HC12	E	Yes	Yes	b473d367d5
HC13	E	Yes	Yes	de7bd69caa
HC14	E	Yes	Yes	649ceaa638
HC15	E	Yes	Yes	649ceaa638
HC16	E	Yes	Yes	6cde492527
HC17	E	Yes	Yes	6cde492527
HC18	E	Yes	Yes	6cde492527
HC19	E	Yes	Yes	6cde492527
HC20	E	Yes	Yes	3016590a80
HC21	E	Yes	Yes	3016590a80
HC22	E	Yes	Yes	3016590a80
HC23	A	Yes	No	3016590a80
HC24	E	Yes	Yes	54886b022f
HC25	E	Yes	Yes	54886b022f
HC26	E	Yes	Yes	0376bdbbe12
HC27	S	No	No	d7913db869
HC28	E	Yes	Yes	a8edee6900
HC29	E	Yes	Yes	a8edee6900
HC30	A	Yes	No	a8edee6900
HC31	E	Yes	Yes	754729ffe5
HC32	A	Yes	No	754729ffe5
HC33	A	Yes	No	754729ffe5
HC34	E	Yes	Yes	754729ffe5
HC35	A	Yes	No	754729ffe5
HC36	A	Yes	No	754729ffe5
HC37	E	Yes	Yes	d976c37825
HC38	A	Yes	No	3c7e243264
HC39	E	Yes	Yes	3c7e243264
HC40	E	Yes	Yes	3c7e243264
HC41	A	Yes	No	3c7e243264
HC42	E	Yes	Yes	446569c893

Notes: E = literal notice text; A = exact address or unambiguous alias; S = secondary source only; U = unresolved. The preferred frame retains 31 E and 9 A records; the literal-notice sensitivity retains only E. The two excluded candidates remain visible. The full UTF-8 crosswalk—including community and school names, matching phrases, source URLs, rationales, local image names, and complete hashes—is `Replication/control_assignment_audit_2017.csv`.

The external formerly-key classification also has independent pre-announcement market content. A school component estimated from 2015–2016 transaction prices improves 2017 out-of-sample prediction, and the

seven formerly-key schools have a higher component than the fourteen ordinary schools. Table A6 reports this treatment-proxy validation. A separately frozen continuous-dose policy specification fails its promotion gates and is not used to rescue the binary result.

Table A6: Pre-announcement market-value validation of the binary exposure proxy

Validation quantity	Result
Historical schools with 2015–2016 support	21 / 21
Schools with at least five support transactions	18 / 21
2017 transaction-level MSE reduction from school component	4.67%
2017 school-mean validation slope ($N = 20$ schools)	0.934
One-sided validation-slope p -value	0.0000062
Formerly-key minus ordinary school-value dose	0.851 SD
One-sided enumerated classification-label sensitivity p -value	0.0462
Continuous policy-dose promotion gate	Fails

Notes: The school component is estimated only from 2015–2016 prices and validated on 2017 transactions before the policy outcome is opened. All twenty-one schools have training support; the validation-slope regression uses the twenty schools with 2017 transactions. The binary key-school classification is external to this exercise. The one-sided label sensitivity enumerates all 116,280 assignments of seven labels among twenty-one schools and is a treatment-proxy diagnostic, not randomization inference for the policy effect. The separately frozen continuous-dose policy interaction fails significance and placebo-dominance gates and is not used as an outcome specification.

Table A7 records the observation counts, communities, clusters, and identifying variation for the headline and supporting samples.

Table A7: Estimation samples and identifying support

	Preferred assignment frame	Exact-notice sensitivity
Completed transactions in four estimation windows	1,173	1,029
Exposed communities	19	19
Comparison communities	40	31
Matched sets	19	19
Historically assigned-school clusters	21	21
Exposed-school clusters	7	7
Community–experiment–side cells	236	200

Notes: The preferred frame retains comparison communities named in contemporaneous 2017 notices or linked by an exact, unambiguous address alias. The exact-notice sensitivity retains only literal text matches. Both samples preserve all nineteen exposed communities and seven exposed schools. Each community contributes four observations: before and after January 1 in the 2018 seasonal comparison and the 2019 policy experiment.

C Additional Identification and Robustness Evidence

Before reconstructing missing completion dates, I recorded two designs in timestamped internal protocols: an announcement-period acceleration and a sixteen-week cutoff comparison. These were not externally preregistered. The announcement estimate provides no evidence of the predicted acceleration. The cutoff estimate has the predicted sign at both endpoints and under both notice restrictions. It nevertheless failed the original promotion gate: only eleven of twenty endpoint-specific window and donut estimates have

wild-cluster $p < 0.05$; the upper-endpoint pseudo-cutoff rank is 0.143 rather than below 0.10; and 27 of 2,515 reconstructed dates (1.07 percent) change cells across endpoints, exceeding a denominator-specific .5 percent ambiguity threshold even though they are only 0.36 percent of all matched transactions. An earlier exact-date screen had already shown the local pattern, so subsequent reconstruction and notice restriction validate a known secondary signal.

Two later additions follow the same discipline. First, a conditional-outcome protocol frozen on August 27, 2026, before inspection, fixed three outcomes on the cutoff cells (log one-plus marketing duration, hedonic log transacted unit price, and the log list-to-transacted ratio), their predicted signs, and restricted-null wild-school inference. Table A21 below re-evaluates those unchanged specifications on the headline notice-verified frame so that the flow and conditional panels share one sample. On the original sixty-one-community frame, the lower- and upper-date estimates were +0.733 and +0.775 for duration, -0.011 and -0.015 for price, and -0.012 at both endpoints for the price ratio. Second, a geographic displacement diagnostic was pre-specified on August 29, 2026 with predictions and all design choices (nearest-exposed distance, the below-median split, and the bizcircle adjacency notion) fixed from geometry alone before any outcome was estimated; all four registered tests are reported regardless of outcome.

The raw 2019 cells are essential for interpreting that sign: completed counts are approximately flat in exposed communities and rise sharply in ordinary-school communities. The cutoff coefficient is therefore a relative DDD shortfall, not an observed collapse in exposed transactions or a districtwide volume estimate. Table A8 reports the full post-protocol geographic anatomy. The four rows are mutually exclusive archive segments; the broader rows use raw totals and are neither assignment-verified controls nor untreated-market estimates. The observed rise is descriptive, not a predicted rebound or an identifying assumption. The absence of a broad Haidian grace-period response also differs from Xicheng, where Shao et al. (2023) report an announcement-period increase followed by an enforcement-period decline in transaction volume.

Table A8: Geographic anatomy of the winter-to-spring rise

Market	Lower completion date			Upper completion date		
	2018 A/B	2019 A/B	Ratio/ratio	2018 A/B	2019 A/B	Ratio/ratio
Formerly key	1.481	0.959	0.647	1.642	1.079	0.657
Ordinary	1.088	1.671	1.535	1.260	1.792	1.422
Other Haidian	1.361	1.199	0.881	1.495	1.332	0.891
Beijing outside Haidian	1.358	1.293	0.953	1.491	1.474	0.988

Notes: A/B is the after-to-before ratio in fixed sixteen-week windows around January 1. Ratio/ratio divides the 2019 A/B ratio by the identical 2018 calendar ratio. Formerly key and ordinary rows use inverse-reuse weighted mean transactions per community on the complete 59-community grid. Other Haidian (excluding that frame) and Beijing outside Haidian use raw archive totals. The groups are mutually exclusive segments from the same archive; broader groups are not assignment-verified controls. The table does not identify substitution or displaced transactions.

The main paper reports the preferred and literal-notice estimates with CR2 intervals and the exposed-school concentration. School sign counts are descriptive concentration diagnostics because controls are reused across matched sets.

The outcome-blind spatial match pairs all nineteen exposed communities to distinct notice-verified controls within 2.384 kilometers. School-equal post-cutoff effects are -0.718 and -0.728 . Under a school-effect symmetry restriction, two-sided sign-flip sensitivities are 0.047 and 0.078; exact-notice controls produce 0.047 at both endpoints. These are not randomization-inference tests. Six of seven school effects are negative,

every exposed-school omission preserves the negative sign, and symmetric Lunar-New-Year exclusions retain 78–96 percent of the corresponding magnitude. The 1.5 km rematch retains only twelve communities and six schools and is near zero; the 2.0 km match retains seventeen, and full support begins at 2.5 km. Baseline housing attributes are balanced, but the 2017 completed-flow standardized difference is about 0.40. The match is therefore corroboration of a local counterfactual, not a second independently decisive design.

Table A9: Outcome-blind spatial-pair validation

Specification	Lower completion date			Upper completion date			Support
	Estimate	Exact p	CR1 p	Estimate	Exact p	CR1 p	Pairs/schools
Exact/alias, 1.0 km	−0.879	0.188	0.173	−1.065	0.125	0.089	7/5
Exact/alias, 1.5 km	0.040	0.938	0.926	−0.065	0.906	0.883	12/6
Exact/alias, 2.0 km	−0.929**	0.047	0.066	−0.813**	0.047	0.077	17/7
Exact/alias, 2.5 km	−0.718**	0.047	0.108	−0.728*	0.078	0.116	19/7
Exact/alias, 3.0 km (primary)	−0.718**	0.047	0.108	−0.728*	0.078	0.116	19/7
Exact-notice controls, 3.0 km	−0.836**	0.047	0.077	−0.845**	0.047	0.078	19/7
Nearest with replacement, 3.0 km	−0.871**	0.016	0.045	−0.752*	0.094	0.121	19/7
LNY blackout ± 14 days	−0.564	0.141	0.150	−0.565	0.172	0.178	19/7
LNY blackout ± 21 days	−0.630	0.203	0.184	−0.643	0.219	0.225	19/7
LNY blackout ± 28 days	−0.692	0.203	0.176	−0.693	0.188	0.223	19/7
Pre-policy placebo: 2018 versus 2017	0.459	0.281	0.323	0.470	0.344	0.365	19/7

Notes: Negative estimates use the paper’s convention and denote an exposed-relative completed-flow shortfall. The match is maximum-cardinality, minimum-total-distance and uses only coordinates and the independent 2017 notice audit; except where stated, controls are distinct. The primary 3 km assignment was frozen before matched outcomes were opened. Estimates first average community-pair DDDs within exposed school and then average schools. Exact two-sided p -values enumerate all 2^G joint sign flips at the exposed-school level; stars use these values: *** $p < .01$, ** $p < .05$, * $p < .10$. CR1 p -values use $t(G - 1)$. Each caliper is rematched, so changing support explains why the 1.5 km row is not comparable to a simple trimming exercise. LNY rows symmetrically remove the indicated days around the year-specific Lunar New Year. The placebo compares the January 2018 differential with January 2017.

Table A10: Balance in the frozen one-to-one spatial match

Predetermined variable	Standardized mean difference
Longitude	−0.244
Latitude	0.037
2017 asinh completed flow, lower date	0.404
2017 asinh completed flow, upper date	0.399
Median log unit price	0.048
Median floor area	0.032
Median construction year	0.117

Notes: Differences are formerly-key minus ordinary, standardized by the pooled dispersion in the nineteen frozen pairs. Housing attributes are available for eighteen pairs. Matching uses coordinates and assignment provenance, not outcomes or these balance statistics. The approximately 0.40 pre-policy-flow difference motivates the later preperiod-balancing diagnostic and remains a limitation of the spatial comparison.

The full-sample surface exercise permits the experiment-side movement to vary smoothly across Haidian. Linear surfaces leave the coefficients slightly larger in magnitude and below 0.05 under school-wild inference at both endpoints. Quadratic surfaces preserve the magnitudes but widen school-wild p -values to 0.105 and 0.138. Conley standard errors are reported only as residual-dependence diagnostics: residual Moran statistics are close to zero and the Conley standard errors fall as the distance cutoff expands, so they do not supply

independent identifying variation.

Table A11: Spatial robustness of the completed-flow triple difference

Specification	Coefficient	School-wild p	Conley p : 1/2/3 km
<i>Panel A. Lower completion date, full preferred sample</i>			
No spatial surface	−0.918** (0.335)	0.012	0.015/0.007/0.003
Linear geography × cells	−0.972** (0.372)	0.021	0.021/0.009/0.004
Quadratic geography × cells	−0.905 (0.432)	0.105	0.026/0.005/0.001
<i>Panel B. Upper completion date, full preferred sample</i>			
No spatial surface	−0.762** (0.376)	0.035	0.057/0.039/0.030
Linear geography × cells	−0.839** (0.395)	0.049	0.053/0.030/0.021
Quadratic geography × cells	−0.766 (0.448)	0.138	0.070/0.031/0.018
<i>Panel C. Previously locked outcome-blind 3 km spatial match</i>			
Lower completion date	−0.718** (0.380)	0.047	—
Upper completion date	−0.728* (0.397)	0.078	—

Notes: The outcome is the inverse-hyperbolic-sine of completed transactions per community in a 16-week cell. Every coefficient uses the paper’s After × 2019 × Formerly-key convention, so a negative value is an exposed-relative, seasonally netted shortfall. Panels A and B retain all 59 communities and the original inverse-reuse weights. All models contain community fixed effects, experiment-by-side fixed effects, and lower-order exposure interactions. Linear and quadratic models permit the stated coordinate surface to differ in every experiment-side cell. Parentheses contain assigned-school CR1 standard errors; stars and the adjacent p -values use 9,999-draw restricted-null school-wild inference. Conley columns report two-sided $t(N - K)$ p -values for Bartlett kernels at 1, 2, and 3 km. Panel C reproduces the previously locked school-equal spatial-pair benchmark; its school-wild column instead reports the exact seven-school sign-flip value. *** $p < .01$, ** $p < .05$, * $p < .10$.

A post-baseline balancing exercise uses the complete local donor universe rather than one control per exposed community. Nonnegative weights are chosen only from January 2015–August 2016 monthly completed flows and coordinates; a 2015 fit and January–August 2016 validation select the penalty before any later outcome enters the final weights. The preferred weights retain all forty controls, have effective sample sizes of 29.6 and 29.8, and put at most 5.6 and 6.4 percent on one control. The coefficients remain negative in the paper’s convention across the complete fixed-penalty family and every recomputed omission of any of the twenty-one represented schools.

The fully held-out 2018-versus-2017 placebo is nevertheless sizeable and opposite-signed, with wild $p = 0.068$ and 0.049 . It is not a clean null, and the cause of the 2017–2018 oscillation is not identified. Multiyear comparisons were added only after observing that adverse result. Relative to the average 2017–2018 seasonal gap, the 2019 estimates remain −0.547 and −0.642; relative to 2017 alone, they are smaller and imprecise. Tables A12 and A13 preserve both the formal result and the adverse diagnostic.

For completeness, Table A14 collects the influence, spatial, and balancing exercises by design status. Figure A2 displays the complete bandwidth/donut family whose specifications were fixed before date reconstruction and all thirteen earlier pseudo-cutoffs. These exhibits are diagnostic rather than an independent research design.

The pre-specified nearby-control diagnostic and its far-control and shared-bizcircle sensitivities are reported in Table A16.

Table A12: Spatially local preperiod-balanced validation

Specification	Lower completion date			Upper completion date		
	Estimate	CR2 p	Wild p	Estimate	CR2 p	Wild p
CV-selected penalty	-0.853**	0.034	0.015	-0.955***	0.028	0.009
Fixed $\lambda = 0$	-0.873***	0.049	0.006	-0.876**	0.094	0.027
Fixed $\lambda = 0.01$	-0.825**	0.045	0.013	-1.003***	0.023	0.004
Fixed $\lambda = 0.1$	-0.853**	0.034	0.015	-0.955***	0.028	0.009
Fixed $\lambda = 1$	-0.877**	0.021	0.011	-0.972***	0.022	0.008
Fixed $\lambda = 10$	-0.881***	0.018	0.009	-0.978***	0.021	0.009
<i>Earlier calendar experiments, CV-selected weights</i>						
2018 versus 2017 (fully held out)	0.611*	0.080	0.068	0.627**	0.062	0.049
2017 versus 2016 (partly in sample)	-0.160	0.581	0.578	-0.056	0.840	0.831

Notes: Negative estimates use the paper's convention and denote an exposed-relative completed-flow shortfall. Control weights are nonnegative, sum to one, and use all notice-verified controls within 3 km of at least one exposed community. The preferred penalty is chosen using January–August 2016 prediction after fitting on 2015; final weights use only January 2015–August 2016 monthly flows and coordinates. CR2 uses coefficient-specific Satterthwaite degrees of freedom. Wild p -values use 9,999 restricted-null draws over every represented exposed and control school; stars use those values. The design was formalized after an exploratory pilot. The fully held-out placebo is opposite-signed and is not a clean pretrend pass. *** $p < .01$, ** $p < .05$, * $p < .10$.

Table A13: Year-specific seasonal gaps and multiyear diagnostics

Year/contrast	Lower completion date			Upper completion date		
	Estimate	CR2 p	Wild p	Estimate	CR2 p	Wild p
<i>Panel A. Formerly-key-minus-ordinary after-minus-before gap by year</i>						
2016	0.015	–	–	0.005	–	–
2017	-0.145	–	–	-0.051	–	–
2018	0.466	–	–	0.576	–	–
2019	-0.387	–	–	-0.380	–	–
<i>Panel B. 2019 relative to alternative seasonal benchmarks</i>						
2019 minus 2018 (formal)	-0.853**	0.034	0.015	-0.955***	0.028	0.009
2019 minus 2017	-0.241	0.345	0.325	-0.329	0.229	0.194
2019 minus average of 2017–2018	-0.547**	0.059	0.036	-0.642**	0.046	0.017
2019 minus average of 2016–2018	-0.499**	0.085	0.049	-0.557**	0.068	0.034

Notes: Negative values use the paper's after-minus-before convention. Panel A reports the weighted formerly-key-minus-ordinary seasonal gap by year. Panel B reports 2019 relative to alternative benchmarks. Only 2019 minus 2018 belongs to the formal protocol. The remaining rows were added after observing the sizeable opposite-signed held-out placebo and must be read as post-protocol diagnostics. Weights remain fixed at the CV-selected preperiod values. Stars use all-school wild-cluster p -values: *** $p < .01$, ** $p < .05$, * $p < .10$.

Table A14: Robustness and influence diagnostics

Diagnostic	Lower date	Upper date
Preferred notice-verified effect	-0.918** [0.014]	-0.762** [0.034]
Exact-notice-only effect	-0.870** [0.021]	-0.715* [0.055]
Unweighted asinh effect	-0.753** [0.027]	-0.715** [0.048]
Leave-one-community/school range	[-1.146, -0.750]	[-0.994, -0.595]
Negative matched-set effects	16/19	16/19
Negative exposed-school effects	7/7	6/7
Recorded window/donut range	[-0.987, -0.274]	[-0.924, -0.029]
Negative window/donut effects	10/10	10/10
Absolute rank of real cutoff among 14 dates	1 of 14	2 of 14
Full sample: linear spatial surface	-0.972** [0.021]	-0.839** [0.049]
Full sample: quadratic spatial surface	-0.905 [0.105]	-0.766 [0.138]
Spatial pairs: 3 km, exact/alias	-0.718** [0.047]	-0.728* [0.078]
Spatial pairs: 3 km, exact notice	-0.836** [0.047]	-0.845** [0.047]
Spatial pairs: pre-policy placebo	0.459 [0.281]	0.470 [0.344]
Local preperiod-balanced controls	-0.853** [0.015]	-0.955*** [0.009]
Balanced held-out placebo: 2018 vs. 2017	0.611* [0.068]	0.627** [0.049]
Balanced 2019 vs. 2017–18 mean (post-protocol)	-0.547** [0.036]	-0.642** [0.017]

Notes: Effects use the main-text sign convention; negative values denote a lower formerly-key-minus-ordinary after-before contrast in 2019 than in the comparison experiment. They do not require formerly key-school counts to fall in levels. The preferred and exact-notice rows use exhaustive restricted-null wild inference over all 2^{21} assigned-school Rademacher sign vectors. Spatial-surface and balanced rows retain their recorded 9,999-draw assigned-school wild p -values; the unweighted row uses its assigned-school CR1 p -value. All original-design diagnostics preserve the original matching when units are omitted. The window/donut family contains seven symmetric windows and three 16-week donut exclusions for each timing endpoint. Rank 1 denotes the largest absolute coefficient among the real January 2019 cutoff and thirteen earlier pseudo-cutoffs. Linear and quadratic surfaces interact centered geography with every experiment-side cell. The separately frozen one-to-one spatial-pair rows use exact two-sided exposed-school sign-flip inference. Balanced-control weights are nonnegative and selected only with January 2015–August 2016 monthly flows and coordinates; their design was formalized after an exploratory pilot. The 2019-versus-2017–18 row was added after the opposite-signed held-out placebo was observed. Stars follow *** $p < .01$, ** $p < .05$, * $p < .10$. School sign counts from the original design describe concentration and are not design-exact inference.

Table A15: Long-horizon persistence and falsification

Window and benchmark	PPML relative-flow ratio		Asinh school-wild p -value	
	Lower	Upper	Lower	Upper
16-week cutoff / identical 2018 calendar	0.422	0.462	0.012	0.035
Calendar 2019 / pooled 2015–2017	0.837	0.843	0.145	0.114
Calendar 2020 / pooled 2015–2017	1.021	1.013	0.871	0.893
Jan. 2019–Dec. 2020 / Jan. 2016–Dec. 2017	0.945	0.945	0.815	0.835
2017 placebo / pooled 2015–2016	0.881	0.887	0.121	0.193

Notes: Ratios below one denote lower formerly-key-relative completed flow. The first row reproduces the main sixteen-week design and exhausts all 2^{21} assigned-school Rademacher sign vectors under the restricted-null, fully studentized wild procedure. The remaining rows come from a separately specified persistence exercise estimated after the core cutoff analysis and retain 9,999 restricted-null draws clustered by historically assigned school. Annual models compare each indicated year with pooled pre-policy years; the two-year model compares cumulative calendar windows. The 2019 annual PPML cluster- t p -values are 0.063 and 0.064; the table reports ratios for scale rather than PPML significance. The similar 2017 placebo limits a causal interpretation of the calendar-2019 result.

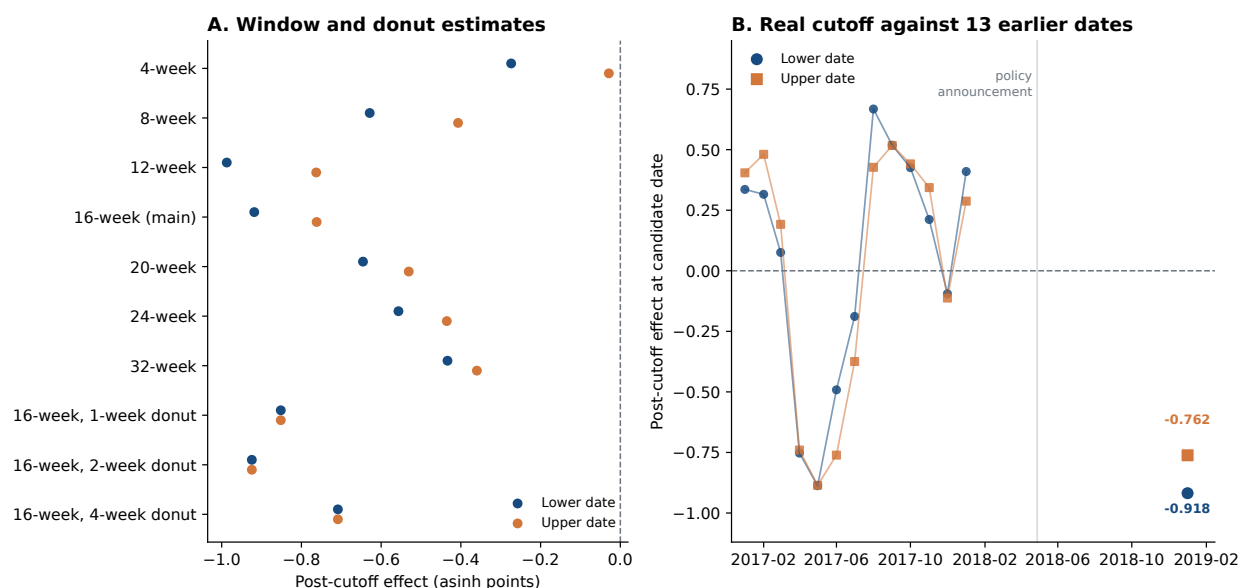


Figure A2: Specification and pseudo-cutoff diagnostics

Notes: Panel A displays every coefficient from the bandwidth/donut specification family fixed before date reconstruction. Panel B displays the real January 2019 coefficient and thirteen earlier pseudo-cutoffs. Coefficients are shown without significance filtering; preferred few-cluster inference appears in the main paper.

Table A16: Geographic displacement diagnostics

	Lower date	Upper date
Near-control extra completed-flow increase (controls only)	-0.322 (0.426)	-0.040 (0.450)
wild-school p	0.393	0.918
Spearman correlation, control increase vs. distance	+0.068	-0.062
Headline DDD, far controls only	-1.003 (0.414)	-0.773 (0.465)
wild-school p ; PPML ratio	0.021; 0.406	0.092; 0.467
Headline DDD, excluding same-bizcircle controls	-0.930 (0.441)	-0.703 (0.483)
wild-school p ; PPML ratio	0.047; 0.417	0.155; 0.473

Notes: Pre-specified diagnostics for displacement of exposed-community transactions into nearby ordinary-school controls. Distance is from each control community to its nearest exposed community; the forty controls sit close to exposed communities (median nearest distance 1.00 km, maximum 2.54 km), so a substitution response should be visible in the near half. “Near” is below-median distance (20 of 40); the first row interacts the near indicator with the full experiment-side structure in the control-only cell panel, so a positive value would indicate an increase concentrated beside exposed communities. The middle row reports the rank correlation between each control community’s own difference-in-differences and its distance. The lower rows re-estimate the headline design keeping only far controls, and excluding the 20 controls that share a bizcircle with an exposed community. Wild-school p -values use the restricted-null studentized procedure with 9,999 draws.

Outcome-blind spatial pairs in Haidian
19 distinct pairs; median 1.32 km; maximum 2.38 km

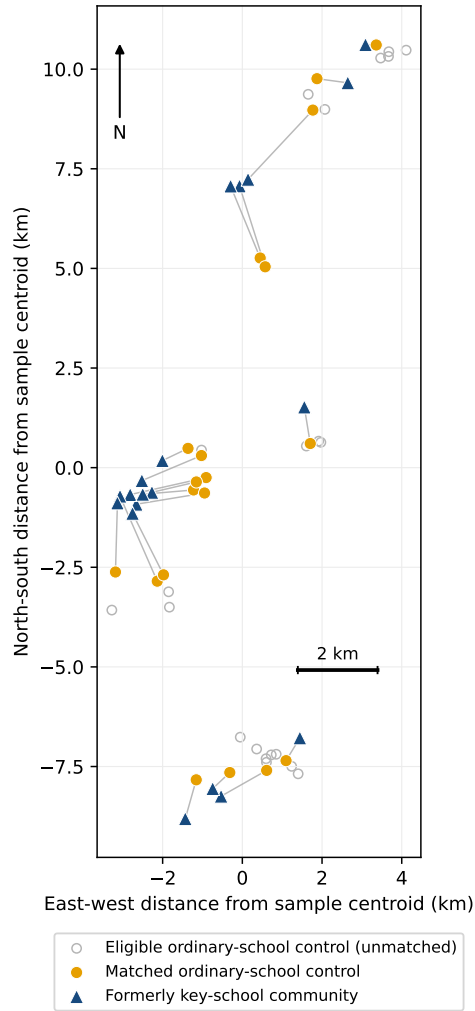


Figure A3: Outcome-blind one-to-one spatial pairs

Notes: Lines connect matched communities; they are not school-catchment boundaries. The global assignment maximizes retained exposed communities and then minimizes total geographic distance, without replacement, using only coordinates and the independent 2017 assignment audit. Coordinates share the same source coordinate system, so pair distances and relative positions are internally comparable.

The matched pairs also permit a descriptive local decomposition. The formerly-key member's share of pair flow shifts by -10.4 and -11.5 percentage points, while combined pair flow shifts by -0.140 and -0.214 asinh points. Neither component is separately precise under the exhaustive seven-school sign-flip sensitivity, which requires school-effect symmetry and is not randomization inference. Because the nineteen pairs do not form a closed market, this diagnostic cannot distinguish local diversion, flows to unmatched locations, and aggregate suppression.

Table A17: Local spatial-pair total and share diagnostics

Outcome	Lower completion date		Upper completion date	
	Coefficient	Exact p	Coefficient	Exact p
Asinh pair-total transactions	-0.140 (0.321)	0.688	-0.214 (0.343)	0.547
Formerly-key transaction share (p.p.)	-10.372 (6.915)	0.172	-11.532 (6.776)	0.125

Notes: The sample is the 19 distinct outcome-blind 3 km pairs. Each coefficient is the 2019 after-minus-before change net of the identical 2018 change, averaged first within treated school and then equally across the seven treated schools. Parentheses report CR1 treated-school standard errors; p -values and stars use the exact two-sided 2^7 school sign-flip distribution. A negative combined-total coefficient is a local pair-level shortfall. A negative share coefficient is a movement away from the formerly key member of the pair. The pairs do not form a closed market, so neither outcome identifies districtwide trade suppression. *** $p < .01$, ** $p < .05$, * $p < .10$.

Table A18: Analysis hierarchy and falsification checks

Check	Lower timing endpoint	Upper timing endpoint	Interpretation
Notice-verified cutoff effect	-0.918** (exact wild p=0.012)	-0.762** (exact wild p=0.035)	Main completed-flow estimate
Exact-notice controls only	-0.870** (exact wild p=0.022)	-0.715* (exact wild p=0.056)	Assignment sensitivity
Monthly pre-period slope (61-community frame)	cluster-t p=0.029; wild p=0.121	cluster-t p=0.059; wild p=0.190	Inference differs with small-cluster correction
Real-cutoff rank among earlier dates	rank p=0.071	rank p=0.143	Only the lower endpoint falls below the 10 percent rank threshold
Announcement-to-cutoff acceleration	-0.380 (wild p=0.308)	-0.336 (wild p=0.392)	Estimate recorded before date repair

Notes: Cutoff effects use the main-table sign, so negative values denote a lower formerly-key-minus-ordinary after-before contrast in 2019 than in the identical 2018 calendar experiment; they do not require formerly key-school counts to decline in levels. Stars use restricted-null wild-cluster inference: *** $p < .01$, ** $p < .05$, * $p < .10$. The first two rows report exhaustive assigned-school sign distributions; other rows retain their named diagnostics. Lower and upper are the two endpoints of the fixed one-day reconstruction interval. Headline and exact-notice rows use the 59- and 50-community frames; rows explicitly labeled 61-community use the original broader frame. In the 414,979-transaction validation sample, all but one reconstructed date is within one day of the reported date. In the broader frame, 27 of 2,515 reconstructed dates (1.1 percent, or 0.36 percent of all 7,410 records) change cutoff-window membership across endpoints. All 20 endpoint-specific bandwidth/donut specifications were fixed before date reconstruction and later produced the predicted sign, but only 11 have wild-cluster $p < .05$. The cutoff was a secondary estimand in timestamped internal protocols, not an external preregistration.

D Completed-Pipeline and Public-Stock Accounting

Let $I_i = 1$ denote that listing i appears as a completed transaction before the archive ends in January 2021. Define selected end-of-day active stock, posting, and exit as

$$S_{ct}^C = \sum_{i:c_i=c} I_i \mathbf{1}\{a_i \leq t < d_i\}, \quad (\text{A7})$$

$$L_{ct}^C = \sum_{i:c_i=c} I_i \mathbf{1}\{a_i = t\}, \quad (\text{A8})$$

$$X_{ct} = \sum_{i:c_i=c} I_i \mathbf{1}\{d_i = t\}. \quad (\text{A9})$$

They obey $S_{ct}^C = S_{c,t-1}^C + L_{ct}^C - X_{ct}$. The implemented identity has zero error in every community-period cell under both date conventions. This stock conditions on completion by the archive end and is not the full offered inventory.

Table A19: Selected observed-completer pipeline accounting

	Lower date	Upper date
Selected-posting PPML ratio	0.962	0.962
Completed-exit PPML ratio	0.422	0.462
Maximum accounting-identity error	0	0

Notes: Posting, stock, and exit are defined only within listings observed to complete by the January 2021 archive end. End-of-day stock obeys $\Delta S^C = L^C - X^C$ exactly in every community-experiment-side cell. Ratios below one denote a weaker formerly-key-relative 2019 after response after subtracting the identical 2018 calendar contrast. These objects do not measure full offered inventory, withdrawal, relisting, later completion, or a marketwide sale hazard.

D.1 Completion timing within fixed posting cohorts

The selected-cohort clearance diagnostic does not survive. It fixes posting cohorts from January 1 through April 22 in 2018 and in the identical 2019 calendar window, then asks whether listings observed to complete by the archive end do so by each of seven horizons. Both cohorts receive at least 540 days of follow-up. At thirty days, the formerly-key-relative 2019 shift is -0.280 and -0.277 under the lower and upper completion dates; nominal CR2 and restricted-null school-wild p -values are approximately 0.04, but no horizon survives Holm adjustment. A one-year-earlier falsification reverses strongly: day-30 completion shifts by $+0.274/+0.271$ and log duration by roughly -0.85 . The 2019 treated cell contains 155 listings from seventeen communities but only five treated schools. Because the sample conditions on observed completion, long-horizon convergence is mechanical and does not imply that all offered homes eventually sell. This exercise supplies no affirmative evidence that affected homes take longer to sell.

Table A20: Clearance within fixed posting cohorts of observed completers

Days	Community-FE DID		CR2 inference		Wild inference	
	Lower	Upper	p_L	p_U	p_L	p_U
30	-0.280	-0.277	0.040	0.041	0.037	0.036
60	-0.281	-0.277	0.205	0.215	0.175	0.190
90	-0.202	-0.202	0.342	0.342	0.407	0.412
112	-0.154	-0.167	0.405	0.396	0.469	0.468
180	-0.046	-0.046	0.704	0.704	0.730	0.731
365	-0.063	-0.063	0.222	0.222	0.244	0.245
540	+0.007	+0.007	0.801	0.801	0.797	0.794

Notes: The sample contains 709 listings posted from January 1 through April 22 in 2018 or 2019 and observed to complete by the January 2021 archive end; both cohorts have at least 540 days of follow-up. The 2019 treated cell has 155 listings from 17 communities but only 5 treated schools. Each coefficient is the formerly-key versus ordinary change in completion-by-horizon probability between the 2018 and 2019 posting cohorts, with community and cohort-year fixed effects. CR2 uses coefficient-specific Satterthwaite degrees of freedom; wild values use 9,999 restricted-null assigned-school draws. At day 30, Holm-adjusted p -values are 0.279 and 0.290 under CR2 and 0.261 and 0.256 under the wild procedure for the lower and upper dates; no horizon survives adjustment across seven horizons. Conditioning on observed completion excludes withdrawals and unresolved or later-completing listings. Long-horizon convergence is mechanical within this selected sample, not evidence that all offered homes eventually sell.

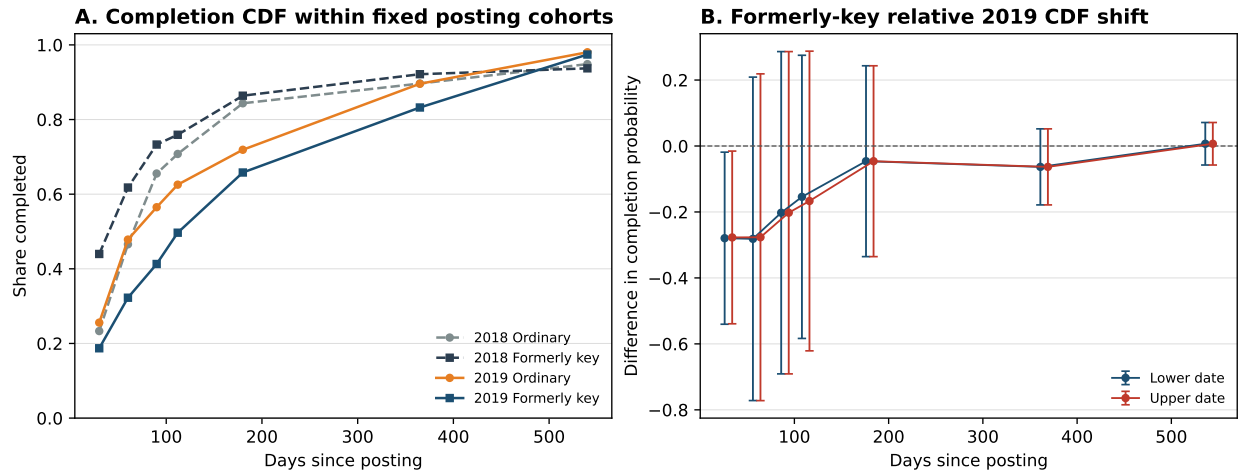


Figure A4: Among observed completers, early completion shifts are negative in point estimates

Notes: Posting cohorts run from January 1 through April 22 in 2018 and 2019, receive at least 540 days of follow-up, and condition on completion by the archive end. Panel A reports design-weighted raw CDFs at the lower completion date. Panel B reports community-fixed-effect difference-in-differences estimates with CR2/Satterthwaite 95 percent intervals under both completion-date conventions. The day-30 unadjusted p -values are approximately 0.04, but no horizon survives Holm adjustment across seven horizons. The 2019 treated cell has 155 listings from seventeen communities and five treated schools. Long-horizon convergence is mechanical in this selected sample, not evidence that all offered homes eventually sell.

D.2 Duration and prices conditional on completed sales

Table A21 reports the complete pre-specified conditional-outcome family on the headline notice-verified frame. Every row conditions on a completed transaction. The duration estimate therefore describes the selected distribution of successful spells, while the price and discount estimates describe selection among trades that clear; none is a full-market sale probability or duration distribution.

Table A21: Incidence around the cutoff: trade, time, and the price of what trades

	Lower date	Upper date
<i>Panel A. Completed transaction flow (community cells, $n = 236$)</i>		
Asinh triple difference	-0.918 (0.335)	-0.762 (0.376)
exhaustive wild-school p	0.012	0.035
PPML completed-flow ratio	0.422	0.462
<i>Panel B. Conditional on a completed sale (transactions, $n = 1,173$)</i>		
Log(1 + marketing duration)	+0.699 (0.422)	+0.745 (0.274)
wild-school p [Holm]	0.238 [0.714]	0.043 [0.130]
Log transacted price per m ²	-0.011 (0.043)	-0.016 (0.041)
wild-school p [Holm]	0.795 [0.856]	0.686 [1.000]
Log list-price-to-transacted-price ratio	-0.013 (0.016)	-0.012 (0.018)
wild-school p [Holm]	0.428 [0.856]	0.534 [1.000]

Notes: All coefficients are exposed-relative after-minus-before effects net of the 2018 seasonal contrast, in the notice-verified preferred frame with headline design weights and assigned-school clustering. Panel A repeats the paper's headline completed-flow design. Panel B applies the pre-specified conditional-outcome models to the same frame: community and experiment-side fixed effects with lower-order exposure interactions; the price and list-to-transacted regressions add log size, construction year, and floor-plan controls. Wild-school p -values use the restricted-null studentized procedure with 9,999 draws; brackets apply a Holm correction across the three Panel B outcomes. Panel B conditions on sale, so these are selected outcomes: under the framework the transacted-price response is not signed, and its null is not evidence about capitalization.

Table A22 reports the full outcome family fixed before the attention coefficients were inspected, together with the subsequent scale-corrected physical-showing-rate audit. The duration component passed that narrow protocol's sign and inference criterion, but it does not survive the broader endpoint-conservative mechanism family and the fixed-cohort prior-year diagnostic reverses. The joint mechanism-promotion gate also fails because corrected daily-rate noninferiority does not. Duration and cumulative showings are therefore exposure-loaded selected descriptions, not mechanism evidence.

A broader post-draft audit asks whether initial asking prices or recorded price revisions reveal more intensive repricing among the 1,173 listings observed to complete in the four main windows. Table A23 reports the revision outcomes together with duration for scale. Revision incidence and cumulative revision counts rise at the point estimates, but revision intensity per exposure day does not. No outcome survives adjustment across the nine-outcome family. Because every row conditions on eventual completion, these results neither recover behavior among withdrawn or unresolved listings nor identify a full-market listing hazard.

Table A22: Selected pipeline and pre-specified attention outcomes

Panel A. Entry and exit in the completed-listing pipeline

	Lower date	Upper date
Selected-entry PPML ratio	0.962	0.962
Completed-exit PPML ratio	0.422	0.462
Posting events / completed exits	1,364 / 1,173	
Community-experiment-side cells	236	

Panel B. Frozen exact-linked attention and duration outcomes

Outcome, log-one-plus scale	DDD	CR2 p	Wild p	N / clusters
Physical showings per day	-0.069	0.432	0.402	855 / 19
Followers per day	0.064	0.584	0.551	855 / 19
Page views per day	0.148	0.739	0.716	755 / 17
Cumulative physical showings	0.854	0.064	0.040	855 / 19
Marketing duration	0.857	0.035	0.019	855 / 19

Panel C. Scale-corrected physical-showing rate audit

Selected-completer PPML rate ratio	0.728
Two-sided school-clustered p -value against one	0.497
One-sided 95 percent lower rate-ratio bound	0.328
Wild one-sided p -value for $H_0 : RR \leq 0.75$	0.533
Predeclared 75 percent noninferiority gate	Fails

Notes: Every panel conditions on completion observed by the archive end and therefore does not measure the offered-listing universe. Panel B reports the complete frozen outcome family; coefficients are exposed-relative post effects after subtracting the identical 2018 calendar contrast. Wild tests use 9,999 restricted-null draws. The 855-record support contains 18 exposed and 38 control communities assigned to 6 exposed and 13 control schools; the page-view outcome has narrower support. Cumulative showings can rise mechanically with duration. Panel C uses a count model with listing days as exposure and supersedes an invalid noninferiority interpretation of the transformed showings-per-day outcome.

Table A23: Post-draft listing-behavior audit among observed completers

Outcome	Lower completion date Estimate [wild p]	Upper completion date Estimate [wild p]
Log initial asking price	-0.024 [.577]	-0.028 [.497]
Any recorded price revision	+0.267 [.111]	+0.282 [.079]
Log one plus revision count	+0.176 [.140]	+0.200 [.071]
Log one plus revisions per day	-0.012 [.298]	-0.014 [.214]
Revision-rate PPML ratio	1.111 [.796]	1.000 [1.000]
Log one plus marketing duration	+0.699 [.237]	+0.745 [.047]

Notes: Every row conditions on appearing as a completed sale before the archive ends. Linear outcomes use the paper's weighted seasonal DDD and assigned-school wild-bootstrap inference; the PPML row uses listing exposure days as an offset and reports the school-clustered p -value. The estimation sample contains 1,173 completed sales. None of the nine outcomes in the expanded selected-listing family survives Holm adjustment; the upper-date duration p -value becomes .421. Initial ask is not independent price evidence because log initial ask equals log transaction price plus the log ask-to-transaction-price ratio. Positive cumulative-revision estimates do not imply faster repricing: revision intensity per exposure day does not rise.

D.3 A frozen selected-duration tail diagnostic

The mean selected-duration result motivated one distributional diagnostic. Before estimating its policy coefficient, I fixed the long-duration threshold at 85 days, the nearest-rank 90th percentile among preferred-frame completed transactions with finite, nonnegative durations whose lower feasible completion date fell in 2015–2016; fixed the same weighted DDD, completion-date endpoints, school clustering, wild and CR2 inference; and required the predicted sign to survive every treated-school omission at both endpoints. Table A24 reports the complete prespecified estimate and inference family. The point estimates are positive under all exposed-school omissions, but the confidence intervals are wide. They do not establish a selected-duration response or a marketwide duration distribution.

Table A24: Selected completed trades shift toward the long-duration tail

	Lower completion date	Upper completion date
After-minus-before DDD (percentage points)	10.2	18.1
CR1S standard error (percentage points)	12.9	9.7
CR1S <i>p</i> -value	0.437	0.077
CR2 standard error (percentage points)	14.7	10.6
CR2 95% interval (percentage points)	[−27.0, 47.4]	[−8.7, 44.9]
School-wild <i>p</i> -value	0.572	0.136
CR2/Satterthwaite <i>p</i> -value	0.516	0.145
Treated-school leave-out range (percentage points)	[8.6, 12.3]	[15.8, 22.6]
Observations	1,173	1,173
Assigned-school clusters	20	20

Notes: Before estimating the policy coefficient, the long-duration threshold was frozen at 85 days, the nearest-rank 90th percentile among 2,171 preferred-frame completed transactions with finite, nonnegative durations whose lower feasible completion date fell in 2015–2016. The outcome is one when recorded marketing duration is at least 85 days. Entries report the after-minus-before form of the weighted seasonal DDD; positive values indicate a larger exposed-relative long-tail share after implementation. CR1S denotes the conventional finite-cluster-corrected school-cluster covariance. The school-wild procedure uses 9,999 restricted-null draws. CR2 uses an identity working model with Satterthwaite degrees of freedom of 5.31 and 5.28. All seven treated-school omissions preserve the positive after-effect at both date endpoints. The sample conditions on completion and does not estimate the duration distribution of all offered listings.

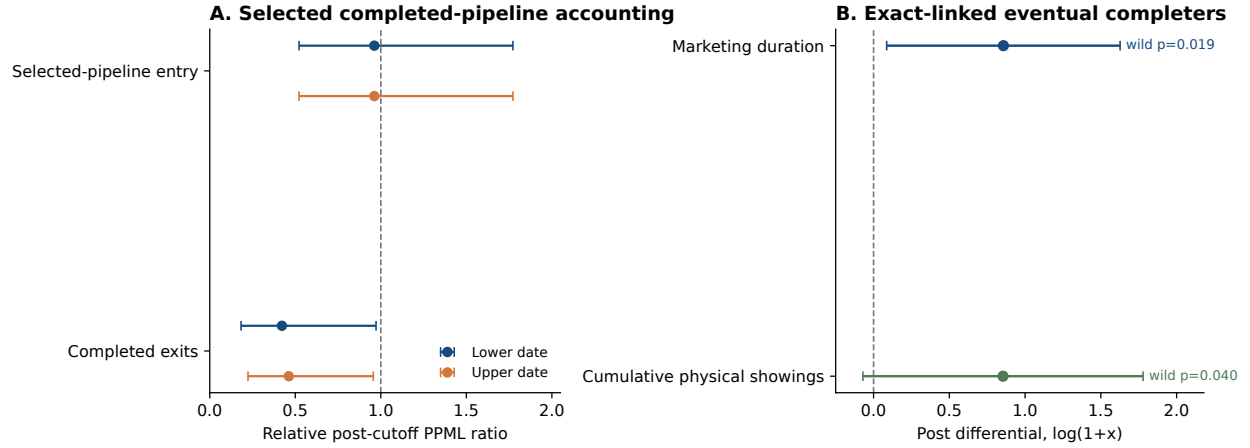


Figure A5: Timing and attention inside the observed-completer pipeline

Notes: Panel A reports exposed-relative PPML ratios for posting into, and exit from, the set of listings observed to complete by the archive end. Panel B reports real-versus-identical-calendar DDD estimates for the unique exact-linked observed-completer sample. Intervals use the displayed equation-specific inference. Neither panel describes the full offered-listing universe.

The offered-stock comparison uses four independent public cross-sections: a November 29, 2017 crawl with exact listing IDs and a source-native community index (Huang, 2017); the H4M real-estate cross-section, whose contents are consistent with a late-March 2018 active-listing wave (Zhao et al., 2022); a broad February 27, 2019 community-by-community crawl with 41,290 active rows across 4,131 named communities; and a November 8, 2019 Lianjia community ledger from a public repository at commit c864b8c. Source-code review confirms iteration over detected community and listing pages in the February file, but it omits listing and dwelling IDs, row timestamps, and a request-success ledger. The 2017 and November-2019 ledgers have stable community IDs and the same Lianjia active-count field across different crawler builds, also without request-success logs. Their union is not a listing panel and does not reveal individual entry, withdrawal, relisting, or terminal states.

Asking-price moments require a valid positive offer in both waves. On these narrower positive-listing supports, the November–February comparison contains 50 communities and the March–February comparison contains 36. Community-mean log asking-price estimates are -0.003 and -0.008 ; median estimates are -0.002 and $+0.003$; and listing-level hedonic estimates are -0.012 and -0.015 . Table A25 reports the full public-listing outcome family. Asking-price dispersion changes sign across baselines, composition results do not cohere across size cutoffs, and nothing survives the within-baseline Holm adjustment. The preferred stock estimates below retain source-native zero-stock communities on broader supports; the positive-listing supports used for price moments are not substitutes for them.

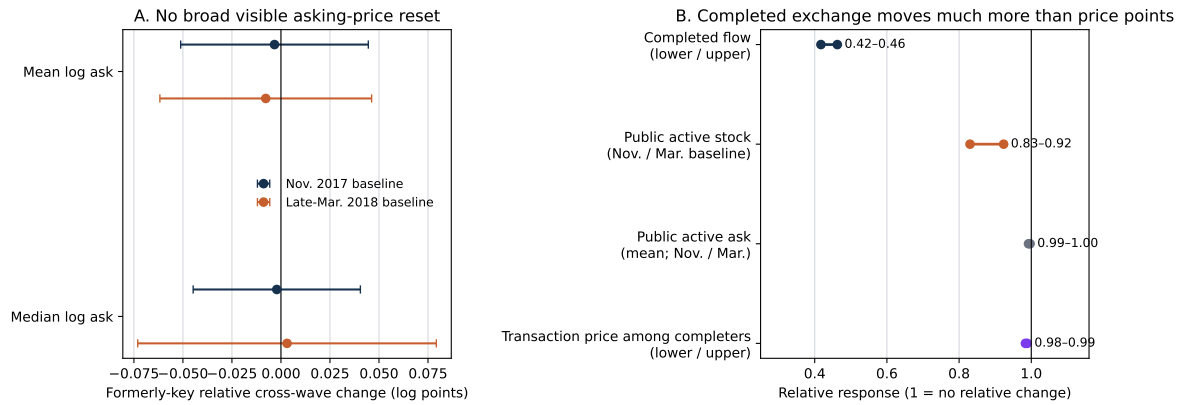
The November-2017–November-2019 exact-ID exercise passes its outcome-blind linkage gate: 18 of 19 exposed communities, 36 of 40 controls, and all seven exposed schools link by unique normalized name and stable Lianjia community ID. Lianjia-visible active listings rise substantially in both arms. The differential asinh change is 0.008 (school-wild $p = .971$), the PPML ratio is 1.059 , and exposed-school omissions cross zero. This is no detectable differential long-run change, not an equivalence result or stable inventory in levels. The 2019 page field labelled total housing units is Lianjia metadata collected after treatment, not cadastral stock. Rate normalization is therefore an influence and scale check, not a new design or a measure of physical turnover.

For the full offered universe, let S_{ct}^O be end-of-period active stock, E_{ct}^O the period's pre-exit exposure to

Table A25: Public active-offer asking prices, dispersion, and composition

Outcome	Nov. 2017–Feb. 2019	Mar. 2018–Feb. 2019
	Estimate [wild p]	Estimate [wild p]
Mean log asking price	−0.003 [.865]	−0.008 [.694]
Median log asking price	−0.002 [.895]	+0.003 [.912]
Hedonic log asking price	−0.012 [.648]	−0.015 [.192]
SD of log asking prices	−0.000 [.990]	−0.012 [.550]
IQR of log asking prices	+0.017 [.564]	−0.047 [.054]
MAD of log asking prices	+0.013 [.257]	−0.015 [.125]
Mean log floor area	−0.115 [.330]	−0.048 [.555]
Share above 60 m ²	−0.098 [.081]	−0.108 [.038]
Share above 90 m ²	+0.039 [.372]	+0.099 [.028]

Notes: Entries are formerly-key-relative cross-wave changes with restricted-null assigned-school wild-bootstrap p -values in brackets. The first two rows use community moments on 50 positive-listing common-support communities for the November baseline and 36 for the March baseline. Hedonic estimates use 958 and 836 listings, community and wave fixed effects, log floor area, and a quadratic in building year, with each community–wave receiving its frozen design weight. Dispersion rows require at least three valid asking prices in both waves and use 35 and 30 communities. No outcome survives Holm adjustment across the nine public-listing outcomes within either baseline. The public waves use different crawlers and do not form a listing-transition panel; dispersion measures posted-price heterogeneity, not latent price uncertainty.



Distinct supports, risk sets, and designs; point estimates are not a decomposition.

Figure A6: Public asking-price points and completed exchange across distinct risk sets

Notes: Panel A reports cross-wave asking-price coefficients with assigned-school-cluster t intervals. Panel B juxtaposes point ratios from different supports and observation universes. Source-native stock ratios retain observed zero-stock communities; price moments require positive valid offers in both waves. The figure is not an additive decomposition, equivalence test, or estimated marketwide hazard.

sale, L_{ct}^O new entries, W_{ct}^O withdrawals, and h_{ct} the execution rate. The accounting system is

$$E_{ct}^O = S_{c,t-1}^O + L_{ct}^O, \quad (\text{A10})$$

$$X_{ct} = E_{ct}^O h_{ct}, \quad (\text{A11})$$

$$S_{ct}^O = E_{ct}^O - X_{ct} - W_{ct}^O. \quad (\text{A12})$$

For aligned exposed-comparison period ratios, this implies $R_F = R_E R_h$, where R_F is relative completed flow, R_E relative period exposure, and R_h the relative execution rate. For baseline source $j \in \{\text{November 2017, H4 March 2018}\}$, the plotted cross-wave stock point is

$$\tilde{R}_{S,j} = \frac{(S_K/S_O)_{\text{February 2019},j}}{(S_K/S_O)_{\text{baseline},j}}, \quad (\text{A13})$$

computed on the identical community support available in source j .

The public snapshots measure a point-stock contrast $\tilde{R}_S$, not period exposure R_E . Define $\kappa \equiv \tilde{R}_S/R_E > 0$ to collect treatment-differential crawler coverage, the unobserved cross-year pre-period stock ratio, and point-snapshot-to-period-exposure alignment. Then

$$R_h = \frac{R_F \kappa}{\tilde{R}_S}. \quad (\text{A14})$$

Proposition A3 (Comparability-indexed point calibration). *If an analyst posits that the combined comparability multiplier belongs to $K = [\underline{\kappa}, \bar{\kappa}]$, then, conditional on point values $(R_F, \tilde{R}_S)$,*

$$\mathcal{H}(K) = \left[\frac{R_F \underline{\kappa}}{\tilde{R}_S}, \frac{R_F \bar{\kappa}}{\tilde{R}_S} \right]. \quad (\text{A15})$$

No execution change is compatible with the points if and only if $\tilde{R}_S/R_F \in K$.

No substantively defended K is estimated from these crawls. The displayed set is therefore an assumption-indexed mapping of point estimates, not a confidence set or an estimated partial-identification region.

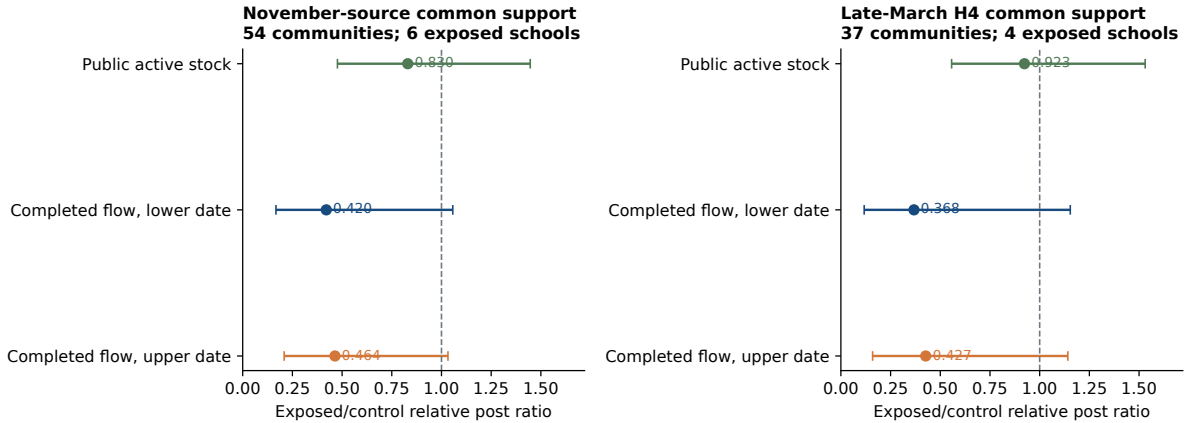


Figure A7: Completed flow and public active stock on identical supports

Notes: Intervals describe equation-specific uncertainty and do not form a joint confidence set for the stock-flow difference. Each comparison uses identical community support within data source.

On the November-source support, the exposed-relative stock point is 0.830 and the exposed-relative completed-flow ratios are 0.420 and 0.464; joint score equality tests give $p = .085/.137$. On the H4 support,

the corresponding stock point is 0.923 and the completed-flow ratios are 0.368 and 0.427; equality gives $p = .029/.033$. Adjusting across the two baseline sources yields a Holm value of .066 for both H4 endpoints. This is a conditional 10-percent survivor, not uniform rejection under every finite-sample correction or identification of a supply-only alternative.

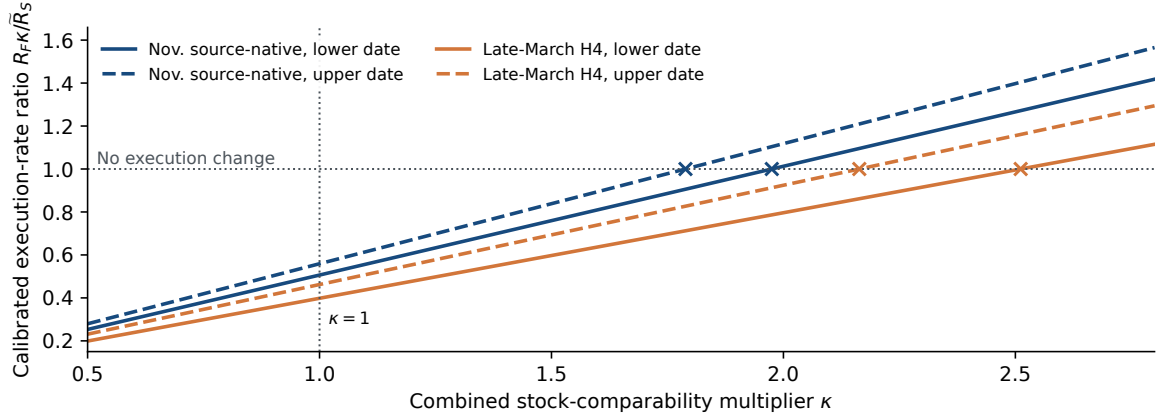


Figure A8: Comparability sensitivity for the execution interpretation

Notes: Curves apply the stock-flow identity to point estimates on identical supports. Crosses mark the comparability multiplier required for no execution change. The figure is not a confidence set and assigns no probability to κ .

At $\kappa = 1$, the points imply execution-rate ratios of approximately 0.40–0.56. A supply-only point fit requires κ between 1.79 and 2.51. The comparison makes the required stock adjustment transparent but does not separately identify buyer arrival, match efficiency, acceptance, withdrawal, or bargaining.

E Post-Headline Extension Audits

This section reports post-headline extensions governed by timestamped internal protocols. They were not externally registered and should not be read as preregistration. Every failed gate and adverse estimate remains in the replication trail.

E.1 External extension audit

I searched the retained archive and public assignment materials for four other Beijing title-vintage rules. Primary-source verification changes the previously proposed Dongcheng date: an official admissions FAQ gives purchasers through December 31, 2018 a certificate grace period, so the legally relevant aggregate boundary is January 1, 2019 rather than June 30. At that corrected boundary, exploratory Dongcheng asinh DDDs are +0.312/+0.467 and PPML ratios 1.678/2.132, opposite the Haidian prediction. The frame fails its frozen assignment and pre-cell-support gates: no control assignment is primary-image verified and exposed pre-side support spans only two positive schools. The adverse estimate is disclosed but is neither a replication nor causal counterevidence.

Xicheng's strict whole-community public map also fails its mapping gate. I therefore ran a separately frozen building-ledger sensitivity. Two outcome-blind reviewers independently transcribed explicit closed building lists and promoted a route only when the listed designators exhausted the Lianjia-recorded building count; their strict intersection was frozen before outcomes. The consensus map contains 12 formerly key

communities in six schools and 20 controls in ten schools and passes the declared cell-support gate. It does not replicate: against the designated 2019 calendar, asinh DDDs are +0.210/+0.126 (exhaustive school-wild $p = .663/.801$) and PPML ratios are 1.020/0.930. Viable older PPML benchmarks range from 0.82 to 1.83 and reverse sign. The aggregate null is also internally unstable by date source: exact-date rows imply acceleration, whereas reconstructed-date rows imply a negative gap; exact-date shares range from 64 to 93 percent across the designated arm–year–side cells. This is a source-format composition warning, not causal heterogeneity. Because the building denominator is post-treatment platform metadata rather than an address registry, this is an exploratory mapping sensitivity, and its failure is not hidden.

Strict public maps also fail for Chaoyang, while Changping supplies only nine post-boundary archive days. Table A26 summarizes the complete feasibility screen. No district is pooled with Haidian, and the paper claims no external replication.

Table A26: Public-data external-extension audit

Candidate	Legal boundary	Empirical result	Frozen-gate verdict
Dongcheng	January 1, 2019	Asinh DDD +0.312/+0.467; PPML ratio 1.678/2.132	Opposite Haidian's predicted sign; fails verified-control-map and pre-cell-support gates
Xicheng	August 1, 2020	Asinh DDD +0.210/+0.126; PPML ratio 1.020/0.930	Strict community map fails; ledger sensitivity passes support but is null, date-format unstable, and reverses across older benchmarks
Chaoyang	July 1, 2017	Not estimated after mapping gate	Thirteen exposed communities span only three exposed schools; four controls span three schools
Changping	January 1, 2021	Not estimable	Archive provides nine, rather than 112, post-boundary days and no verified pre-policy crosswalk

Notes: Legal boundaries come from official district materials. Platform completion dates are only aggregate-response clocks and need not equal the purchase-contract or certificate date that determines legal treatment. Dongcheng's exact school-wild p -values are .272/.108; the upper-date conventional PPML p -value is .005 in the adverse direction. Because its mapping and cell-support gates fail, that estimate is neither a replication nor causal counterevidence. Xicheng's ledger crosswalk was frozen after two independent outcome-blind reviewers agreed on every promoted route; the designated school-wild asinh p -values are .663/.801. Exact-date and reconstructed-date subsets point in opposite directions, so the aggregate null is not a stable source-specific estimate. It does not validate Haidian, and the post-treatment building-count ledger makes it a sensitivity rather than a strict replication. No candidate is pooled with Haidian.

E.2 Exhaustive fixed-weight relabeling over assigned-school clusters

The audit re-estimates three specifications for every one of the $\binom{21}{7} = 116,280$ ways of designating seven of the twenty-one assigned-school clusters as exposed, holding the frozen outcomes, frame, design weights, and fixed effects fixed. For the raw 2019 seasonal DD, the actual grouping ranks 2,427th and 2,589th from the most negative coefficient under the lower and upper clocks. One-sided coefficient tails are .021/.022, two-sided coefficient tails .058/.063, and two-sided CR1-studentized tails .051/.047. The all-pre-mean DDD is weaker: one-sided coefficient tails .022/.050, two-sided coefficient tails .109/.148, and studentized tails .060/.110. These additions show that the directional label sensitivity is not solely a consequence of subtracting 2018.

Table A27 reports the designated-2018 version. Ranked from the most negative coefficient, the actual

grouping is 276th and 1,743rd; ranked by absolute magnitude, it is 2,989th and 5,656th. Descriptive two-sided tail shares are 0.026 and 0.049 on the coefficient and 0.011 and 0.042 on the assigned-school CR1 t -statistic. The 95th percentiles of the absolute relabeled coefficients are 0.816 and 0.757, against absolute observed effects of 0.918 and 0.762; the upper-date comparison is consequently close to that threshold. Key-school status was not assigned, the frame and inverse-reuse weights were constructed around the actual exposure map, and those weights remain fixed. Every exercise is therefore a finite fixed-frame label-sensitivity description, not permutation, design-based, or randomization inference and not an independent confirmation of the wild bootstrap.

Table A27: Exhaustive fixed-weight relabeling over assigned-school clusters

	Lower date	Upper date
Observed after-minus-before DDD	−0.918	−0.762
Label assignments (7 of 21 schools, all subsets)	116,280	116,280
Signed rank (most negative = 1)	276	1743
Absolute rank (largest $ DDD^* = 1$)	2989	5656
Two-sided tail share, coefficient	0.026	0.049
Two-sided tail share, studentized (CR1)	0.011	0.042
Lower-tail share, coefficient	0.002	0.015
95th percentile of $ DDD^* $	0.816	0.757

Notes: For every one of the $\binom{21}{7}$ ways of designating seven of the twenty-one assigned-school clusters as exposed, all communities in those schools are relabeled and the production weighted-asinh DDD is re-estimated with the frozen frame, inverse-reuse weights, community fixed effects, and experiment-side fixed effects. Tail shares describe the finite set of relabelings. Key-school status was not assigned, and the frame and weights were constructed around the actual exposure map and held fixed under relabeling. The reported shares are therefore label-sensitivity summaries, not randomization p -values or an independent inference procedure.

E.3 Unit-type split

The unit-type split reported in the main text recorded two competing readings before estimation: a larger gap in family-sized units, or a larger gap in small units bought mainly for the school right. The 60 square meter cutoff was fixed as a round threshold near the 2019 window’s first quartile; it classifies 262 of the 1,173 observations in the four principal cells as small. The bedroom split uses the archive’s room-count field and omits seven observations with missing room counts. Class-specific estimates re-run the production weighted-asinh design within class; the class difference comes from a pooled model with class-specific fixed effects. The floor-area split satisfies the recorded class-difference and PPML-ordering rule under the designated 2018 benchmark, although its interaction p -values are 0.050–0.052 and older benchmarks attenuate the contrast. The bedroom split does not satisfy the PPML-ordering rule: its PPML ratios are slightly lower for one-bedroom than for two-plus-bedroom units. The evidence is therefore a benchmark-specific heterogeneity diagnostic, not a buyer-family mechanism test.

Table A28 reports the raw 2019 class differences and identical-calendar comparisons against 2016, 2017, and 2018. These comparisons were constructed after inspecting the designated-benchmark heterogeneity. The raw 2019 floor-area interactions are small and imprecise, and none of the floor-area or bedroom interactions survives adjustment across the four benchmark comparisons within an endpoint. This post-estimation

sensitivity table is reported to show exactly where the designated split obtains its apparent strength.

Table A28: Unit-type sensitivity across seasonal benchmarks

Comparison	Small-unit estimate	Large-unit estimate	Small minus large	Wild p	Holm p
<i>Panel A. Floor area ($\leq 60 m^2$ versus $> 60 m^2$)</i>					
Lower completion date					
Raw 2019 DD	−0.333 (0.235)	−0.457 (0.215)	+0.124 (0.185)	0.437	1.000
2019 minus 2016 DDD	−0.048 (0.330)	−0.181 (0.373)	+0.133 (0.373)	0.682	1.000
2019 minus 2017 DDD	−0.225 (0.260)	−0.508 (0.264)	+0.283 (0.314)	0.345	1.000
2019 minus 2018 DDD (designated)	−0.237 (0.330)	−0.963 (0.318)	+0.726 (0.343)	0.052	0.209
Upper completion date					
Raw 2019 DD	−0.339 (0.234)	−0.339 (0.213)	−0.000 (0.159)	0.998	1.000
2019 minus 2016 DDD	−0.054 (0.331)	−0.063 (0.380)	+0.009 (0.398)	0.979	1.000
2019 minus 2017 DDD	−0.231 (0.259)	−0.472 (0.264)	+0.241 (0.340)	0.438	1.000
2019 minus 2018 DDD (designated)	−0.152 (0.373)	−0.904 (0.333)	+0.752 (0.384)	0.050	0.200
<i>Panel B. Bedrooms (one versus two or more)</i>					
Lower completion date					
Raw 2019 DD	−0.165 (0.131)	−0.412 (0.262)	+0.247 (0.252)	0.233	0.466
2019 minus 2016 DDD	+0.227 (0.273)	−0.141 (0.434)	+0.368 (0.490)	0.459	0.466
2019 minus 2017 DDD	+0.037 (0.195)	−0.561 (0.261)	+0.597 (0.257)	0.034	0.131
2019 minus 2018 DDD (designated)	−0.180 (0.261)	−0.850 (0.297)	+0.669 (0.299)	0.033	0.131
Upper completion date					
Raw 2019 DD	−0.170 (0.131)	−0.299 (0.256)	+0.130 (0.257)	0.482	0.964
2019 minus 2016 DDD	+0.222 (0.274)	−0.028 (0.437)	+0.250 (0.499)	0.623	0.964
2019 minus 2017 DDD	+0.032 (0.195)	−0.531 (0.252)	+0.563 (0.282)	0.036	0.143
2019 minus 2018 DDD (designated)	−0.189 (0.281)	−0.786 (0.313)	+0.596 (0.335)	0.048	0.143

Notes: This post-estimation diagnostic uses the frozen 59-community frame, design weights, 16-week windows, and completion-date conventions. The raw 2019 row is the formerly-key-minus-ordinary after-before difference. Each DDD row subtracts the identical calendar contrast ending in the stated benchmark year. Estimates are weighted inverse-hyperbolic-sine coefficients; CR1 assigned-school standard errors are in parentheses. “Small minus large” is the pooled class interaction, so a positive value means a more negative gap for the large class. Wild-school p -values use 9,999 restricted-null studentized draws. Holm values adjust the four interaction tests within each split and completion-date endpoint. These comparisons were constructed after inspecting the designated-benchmark heterogeneity and are sensitivity diagnostics, not pre-specified tests.

F Reproducibility and Scope of Claims

Hashed inputs, assignment samples, and machine-readable result files are recorded in the analysis protocols. A core exhibit builder regenerates the underlying cutoff estimates and figures, while separate routines generate assignment-stability, road-network, persistence, source-format, homogeneous-clock, public-stock, and external-extension audits. The submission tables are assembled from machine-readable outputs produced by these documented modules. Independent routines reproduce sample construction, reuse weights, regressions, wild-cluster and CR2 inference, aggregation, and leave-one-out estimates. For the four headline and exact-notice asinh coefficients, a standalone audit reproduces the archived 9,999-draw Monte Carlo counts and then exhausts all 2^{21} assigned-school Rademacher sign vectors under the same restricted-null, fully studentized procedure. The reported exhaustive values are exact for that bootstrap sign distribution, not assignment-based randomization p -values. A separate independent implementation reproduces every benchmark-relabeling coefficient and tail share to numerical precision.

The project reproduction trail is split between `paper/manuscript/Replication`, which contains the frozen matched frames, core headline and long-horizon modules, and selected-completer summaries, and the September 3 audit

archive under output/salvage.2026-09-03, which contains source-format and homogeneous-clock diagnostics, exact-linked public-ledger files, and the consensus Xicheng crosswalk and failed-extension results. The main transaction input is proprietary and cannot be redistributed without provider permission. Public crawler files, notice images, source URLs, commit identifiers, and their hashes remain separately auditable.

The interpretation used throughout the paper is:

1. **Identified under parallel differential seasonality and treatment-neutral relative platform capture:** a cutoff-local exposed-versus-ordinary DDD in platform-recorded completed flow, not an absolute exposed decline or districtwide contraction.
2. **Exactly accounted but selected:** entry, exit, and stock among eventual sales.
3. **Conditionally calibrated:** execution as a function of public stock and κ .
4. **Not identified:** full inventory, listing-level sale probability, withdrawal, relisting, buyer arrival, permanent trade destruction, or welfare.

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